# ORGANIZATIONAL NETWORKS

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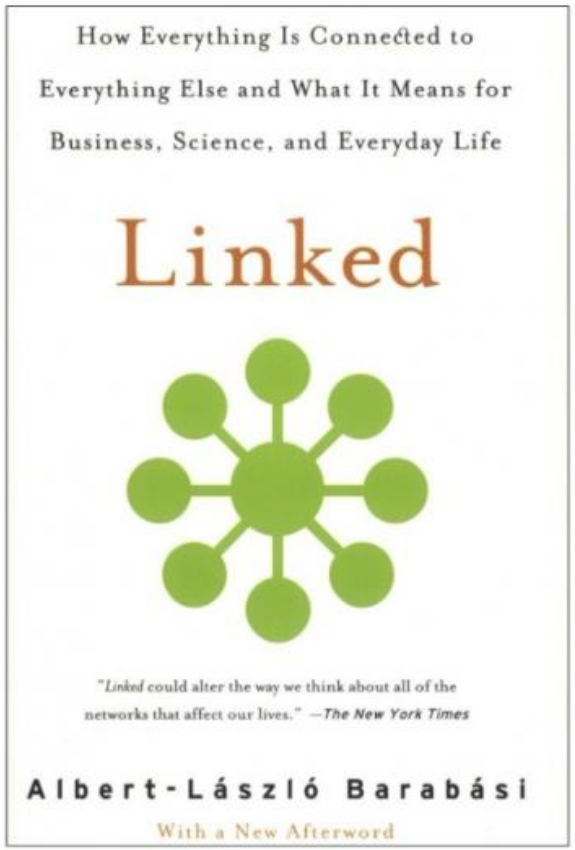
# EXERCISES

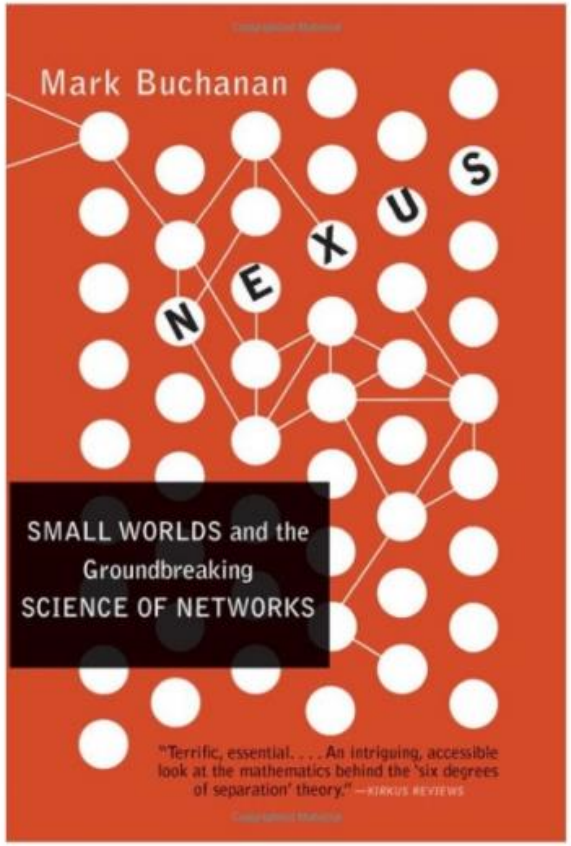
**Tell us why did you choose this course?**

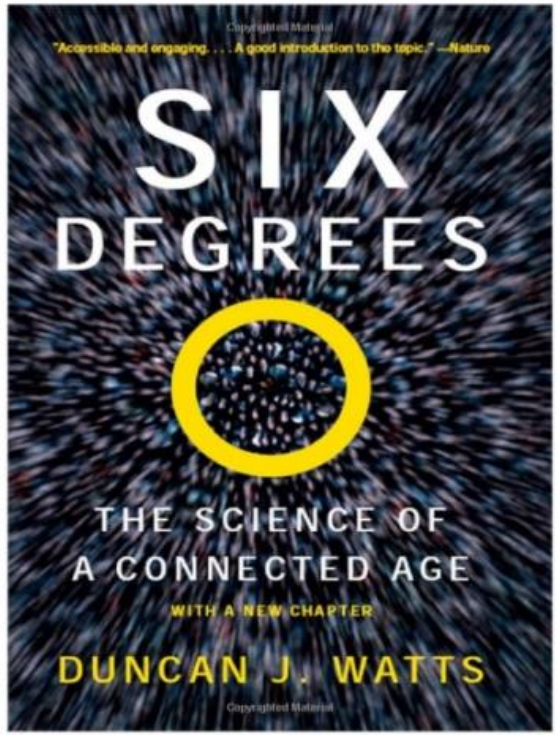
**Did you learn about the graphs or networks?**

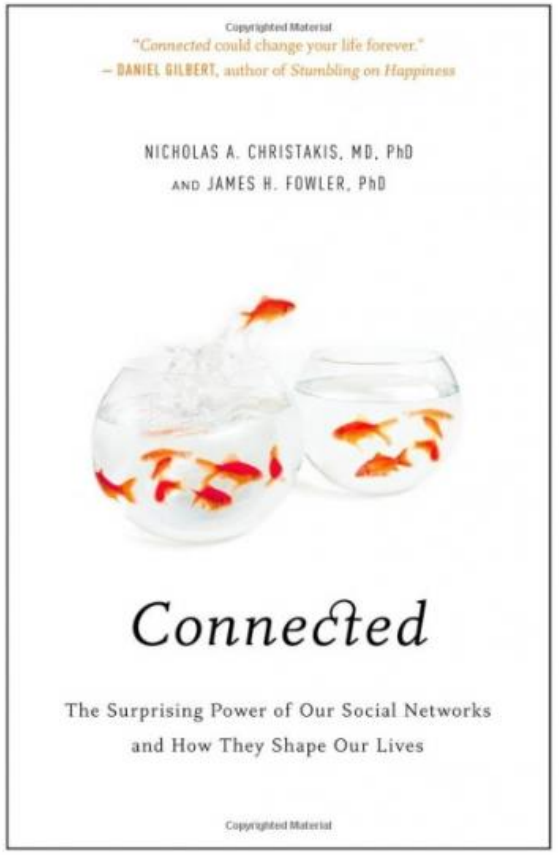
**What is your expectation for this course?**

# LITERATURES



# EXERCISE

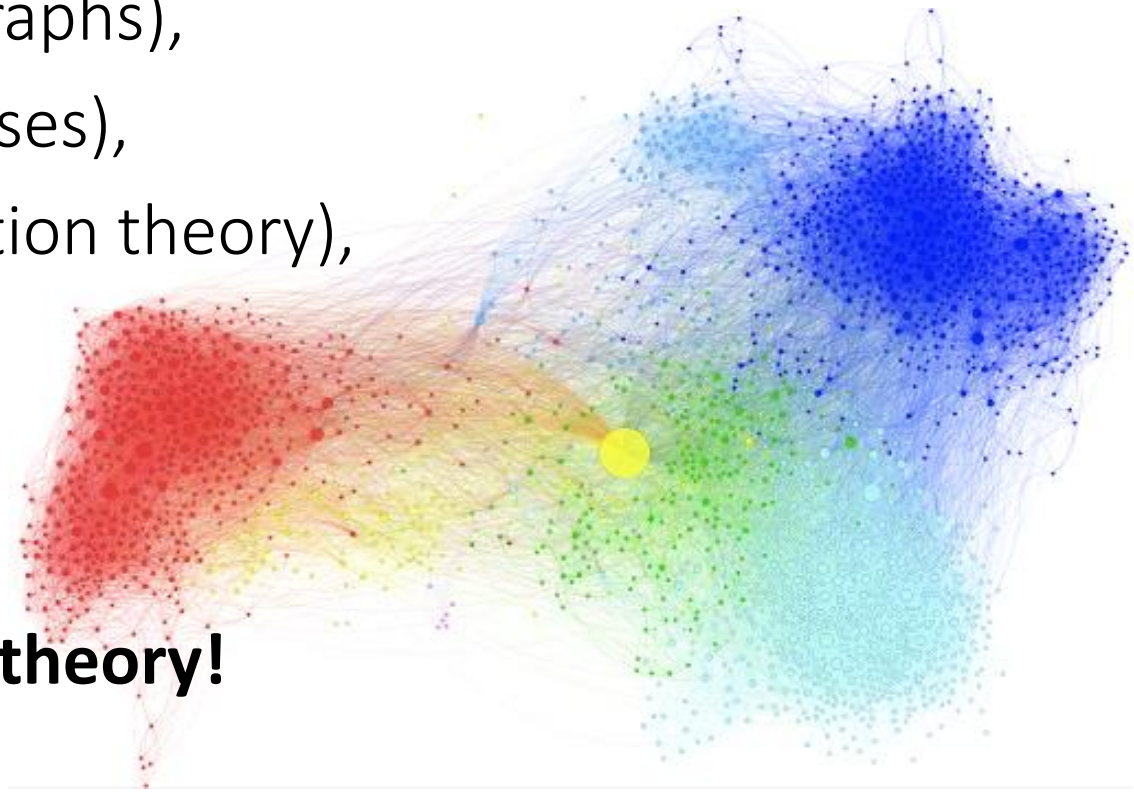
**What means for you the phrase: network?**

**Make notes and tell your answer in some sentences!**

# WHAT THE NETWORK SCIENCE IS?

**This is an interdisciplinary science and it use the methods of**

- the graph theory (properties of the graphs),
- the statistical physics (random processes),
- the engineering (control and information theory),
- the statistic (noisy datasets).



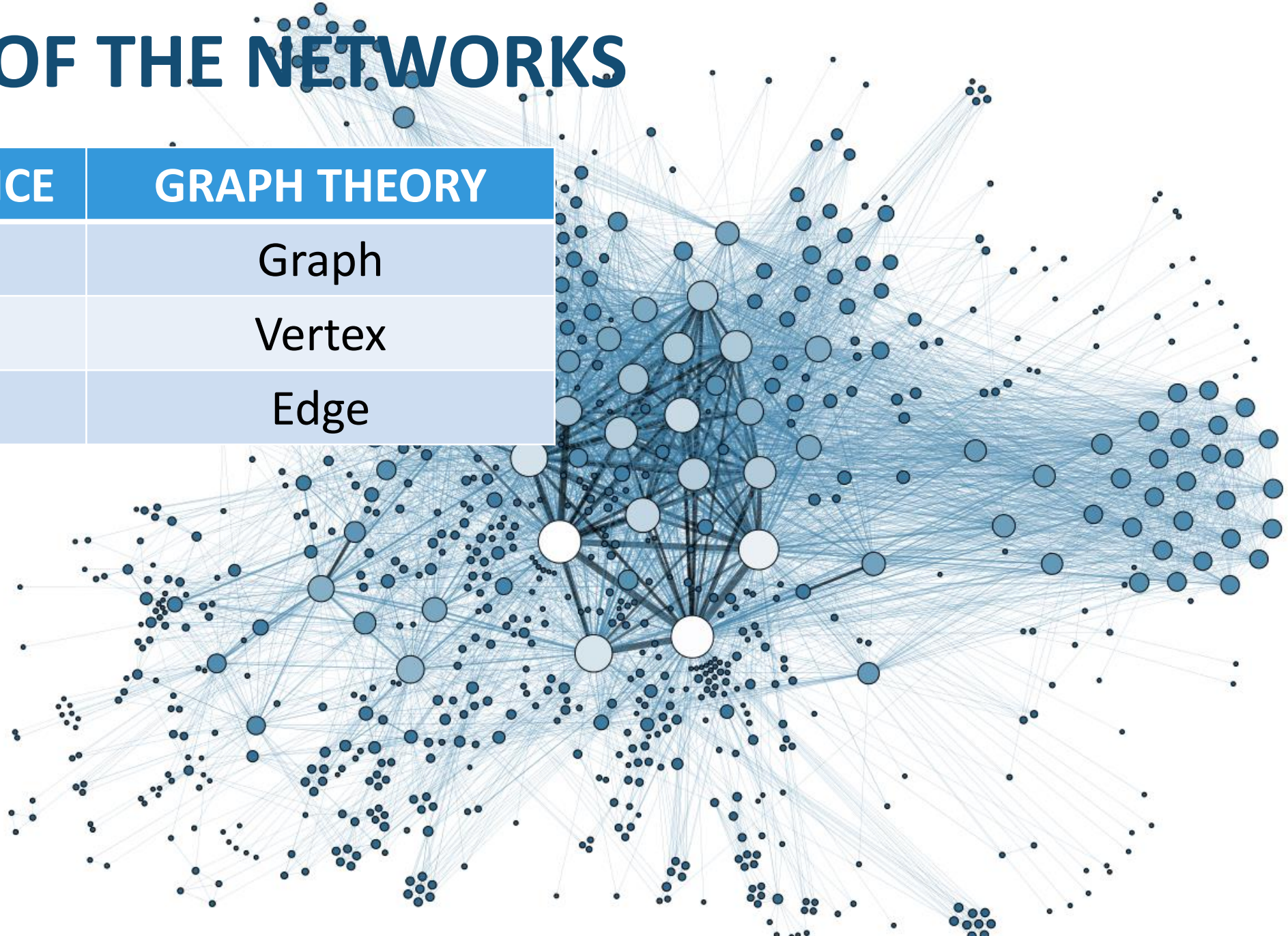
**It’s a young science!**

**This is not the same as that the graph theory!**

We have unimaginable quantity of **data** and we have network analysis **softwares.**

# THE PARTS OF THE NETWORKS

| NETWORK SCIENCE | GRAPH THEORY |
|---|---|
| Network | Graph |
| Node | Vertex |
| Link | Edge |



# DIFFERENT PHRASES

- **social network**
  - sociological connections between people
  - Facebook, Twitter
  - a movie about Mark Zuckerberg
- **human network**
  - social connections
  - organizational connections
  - personal connections
  - compulsionary or voluntary?
- **company network**
  - human connections in a company
  - IT network of the company
- **organizational network**
  - human-machine connections
  - human-human connections

# EXERCISE

**List three different real networks and state the nodes and links for each of them.**

**How many nodes and links could there be in them?**

# EXERCISES

**Please write what do you think:**

**in which areas the network science is used today and for what?**

# WHERE CAN WE USE THE NETWORK SCIENCE?

## Sociology

There are connections between the people:

- friendly connections,
- family connections,
- working connections,
- religious connections,
- residential connections,…



The behavior of an exact person and the behavior of the groups and the mass are different. Why?



# WHERE CAN WE USE THE NETWORK SCIENCE?

## Neuroscience

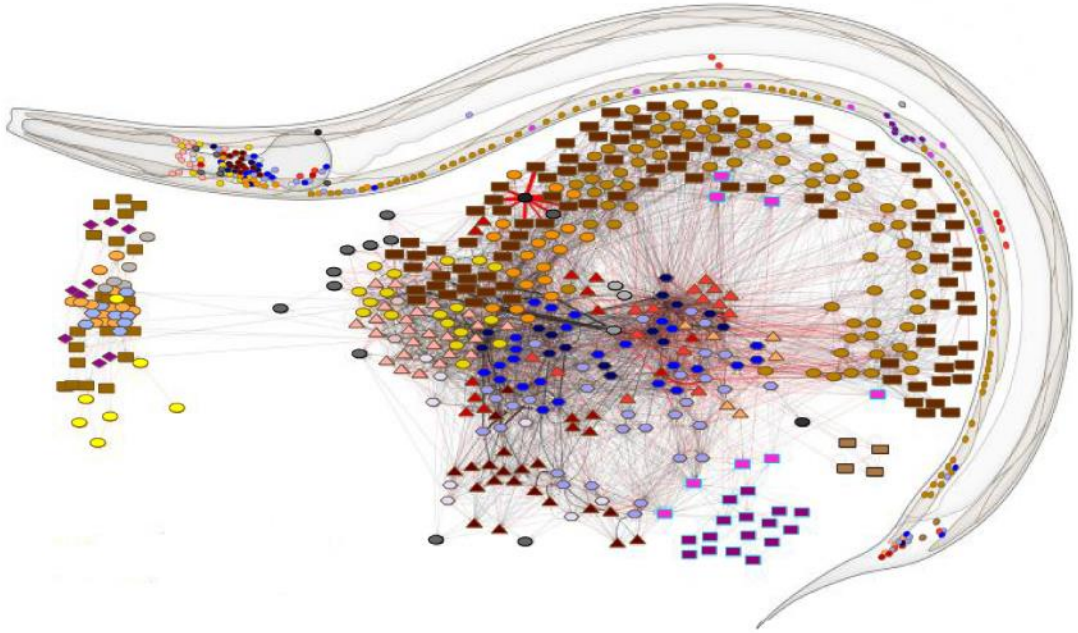
In the human brain are billions of interlinked neurons.

It’s a complex system; neurons are born and die and the links change between the neurons according as how and what we learn and do.

We haven’t a complete map of it.

We have only one complete brain map.

(*C. elegans* worm with 302 neurons)



# WHERE CAN WE USE THE NETWORK SCIENCE?

## Economy

The successful companies built their business strategy on network.

The linked companies are more robustness than the stand-alone ones.

The higher link density gives
higher random failure tolerance
and higher attack tolerance
for a company.



# WHERE CAN WE USE THE NETWORK SCIENCE?

## Health

The first complete list of the human genome at 2001.

But only the list isn’t enough.

We need the map of the cell connections.

The breakdown of the molecular level network
is the responsible for the most human diseases.



# WHERE CAN WE USE THE NETWORK SCIENCE?

## Security

There are network thinking in the law enforcement agencies.

There are network thinking in the military doctrines.

They use network-based weapons and network-based defense tactics.

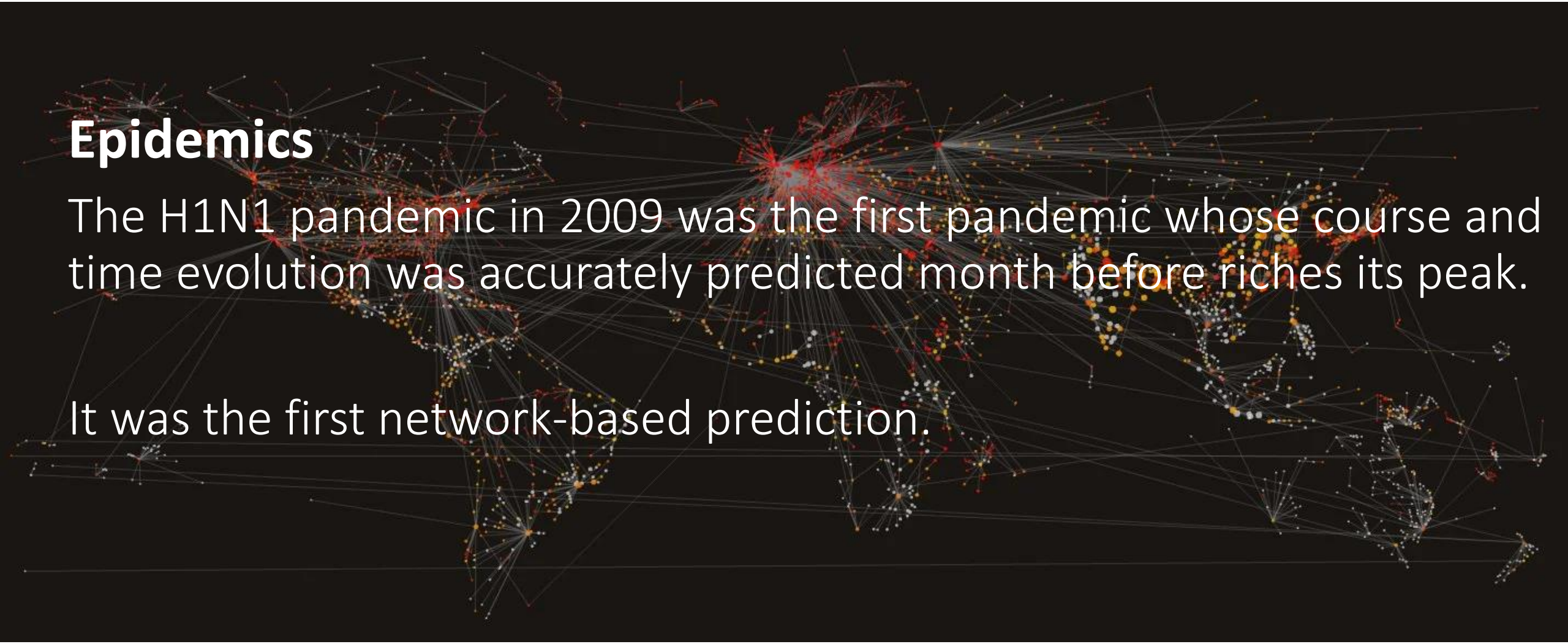
Mapping the terrorist networks.

Police investigations.

They could fight against network by network.



# WHERE CAN WE USE THE NETWORK SCIENCE?



**Epidemics**

The H1N1 pandemic in 2009 was the first pandemic whose course and time evolution was accurately predicted month before riches its peak.

It was the first network-based prediction.

# facebook



# Air traffic



# Air traffic



# EXERCISE

**In your view what would be the area where the network science could have the biggest impact in the next decade?**

**Explain your answer.**

# AN EARLY IDEA



KARINTHY
FRIGYYES

*Chains*

SZÉPIRODALMI KÖNYVKIADÓ

# EARLY SOCIOLOGICAL RESEARCHES

- **Stanley Milgram -** first „small word” experiment.
- There are 2 target persons, with different personal parameters.
- Sent 300 mails with their names and photos to randomly selected persons:
- *„Please try to reach this people whit your friendship links!”*

**The average distance was only**

# 5.5

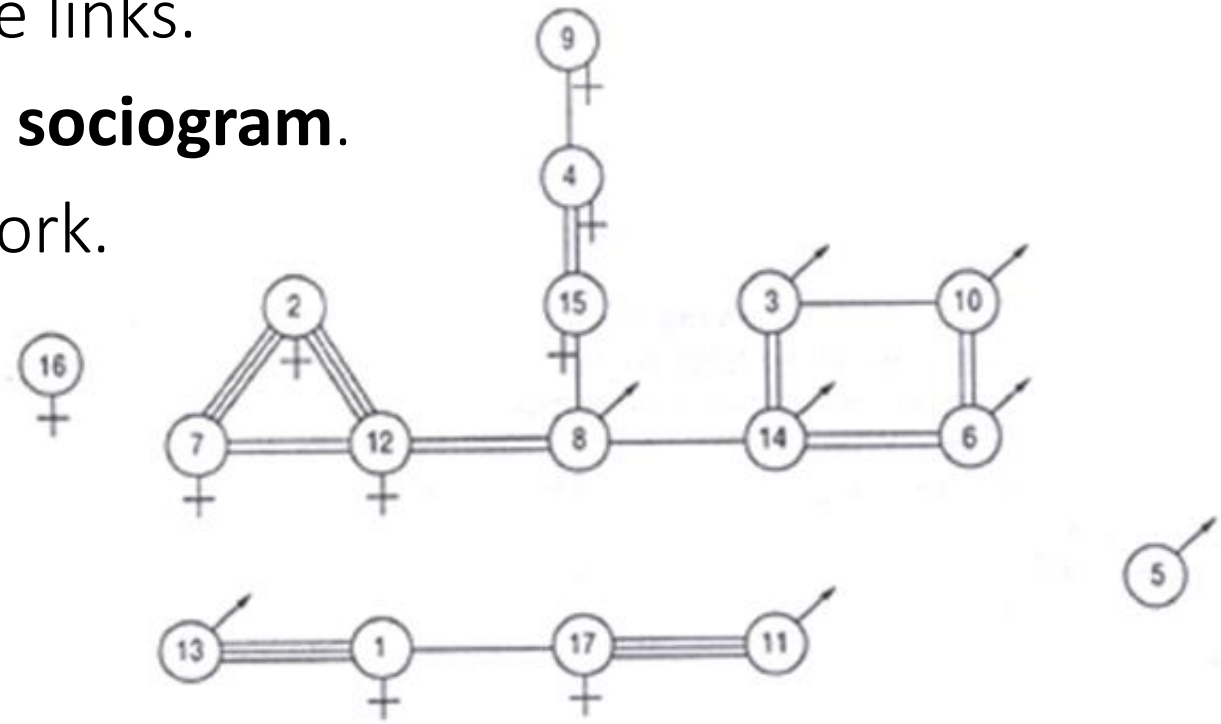
**connections between two unknown persons.**

# EARLY SOCIOLOGICAL RESEARCHES

- **Karinthy Frigyes** wrote a novel: **Láncszemek** (Chains) (1929)
- In 1933 **Jacob Levy Moreno** made a method to figure up the personal connections of the human groups.
- Moreno observed that in the organizations come to be **friendships**, inside groups and intergroup connections.
- These connections forms make a network and **this network is different than the official organizational chart**.
- It was the **sociometry**.
- **Mérei László** naturalized the method of sociometry in Hungary.
- The sociometry **shows only the personal preferences** of the member of the group.

# EARLY SOCIOLOGICAL RESEARCHES

- In the sociometry we use **personal preferences** questions.
- For example: *„With whom you would like to work in a common room?”*
- By the questions have to make a **social matrix**.
- Only the **reciprocal connections** make links.
- By the social matrix have to make the **sociogram**.
- The sociogram is an **undirected** network.



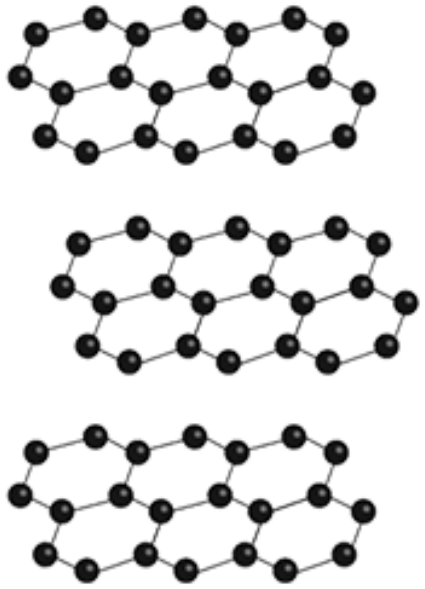
# EXERCISE

**Calculate the Bacon number type distance between two freely chosen Hollywood movie actor!**

- The number must be higher than 2!
- Show the connection chain with movies!

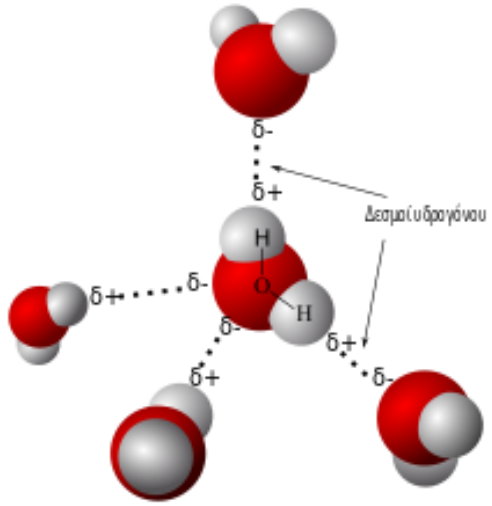
# What is the different?

- **Graphite vs. Diamond?**

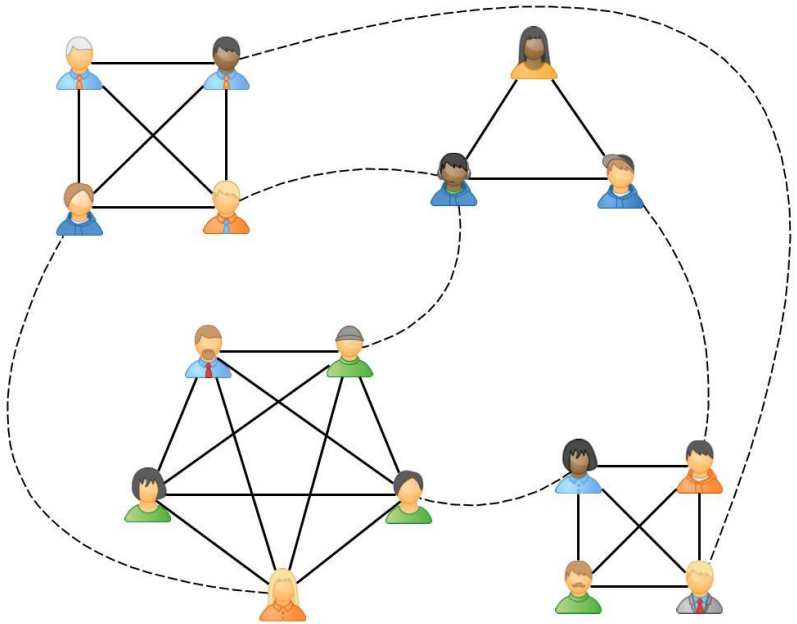




# EARLY SOCIOLOGICAL RESEARCHES




- **Mark Granovetter:** The strength of the weak ties. (1973)
- He focused that how we search and find new job.
- – „Who helped you to find a new job? Was it a friend?”
- – „No, it was not a friend.”
- We use our weak social ties for it.
- We have strong personal connections (ties) with our family and our close friends.
- Our close friends are often friends with each other as well.
- The weak ties are the social bridges.

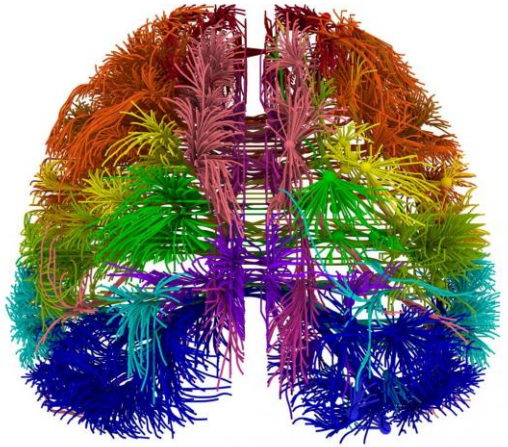


# VIDEO

# EXERCISE

**Describe  your own views on what and how the weak ties can help for a student in his university studies?**

# COMPLEX SYSTEMS



## We are surrounded by complex systems

- **Society**: requires cooperation between billion individuals
- **Communication**: integrates billions of mobile phones, computers
- **Brain**: makes coherent activity of billions of interlinked neurons
- **Biological bodies**: have billions of interactions between metabolites
- **Economy**: integrates countries, governments, people, and billion of companies and organizations
- **Weather**: depend on earth, atmosphere, seas, gravity, temperature…

**The network science can reveal the behaviors of complex system**

# COMPLICATED OR COMPLEX? IT ISN'T SAME!

## COMPLICATED SYSTEMS

- There are **many** parts
- The system **can't be restructured spontaneously**
- There are **some feedbacks** in the system
- Those feedbacks are **planned**
- The parts of the system **don't disappear**
- The behavior of the system **is calculable by the parts**



# COMPLICATED OR COMPLEX? IT ISN'T SAME!

## COMPLEX SYSTEMS

- There are **many** parts
- The system **can be restructured spontaneously**
- There are **more feedbacks** in the system
- The feedbacks **could appear stochastically**
- The system parts **could appear or disappear**
- The behavior of the system **isn't calculable by the parts**



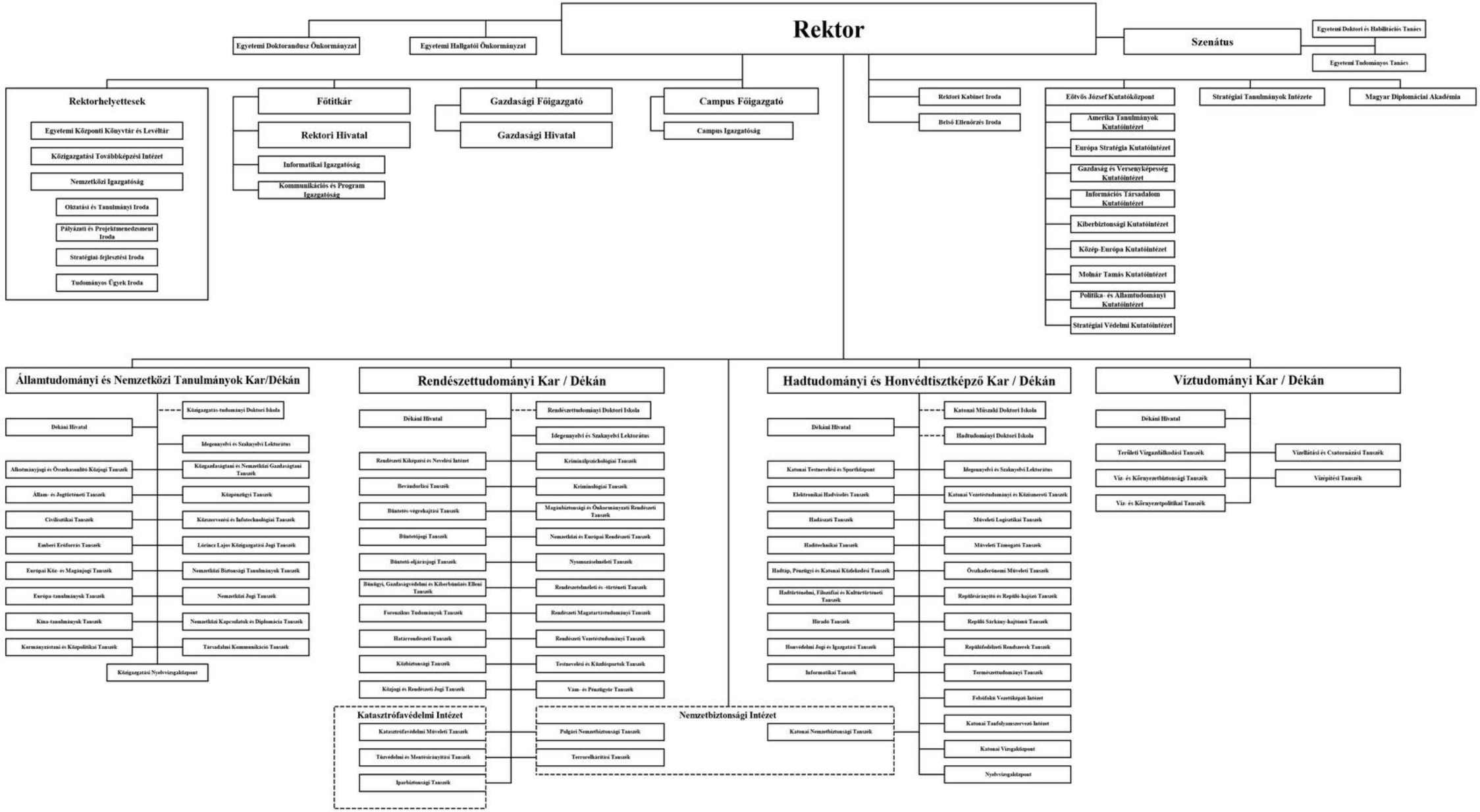
# WHERE CAN WE USE THE NETWORK SCIENCE?

## Management

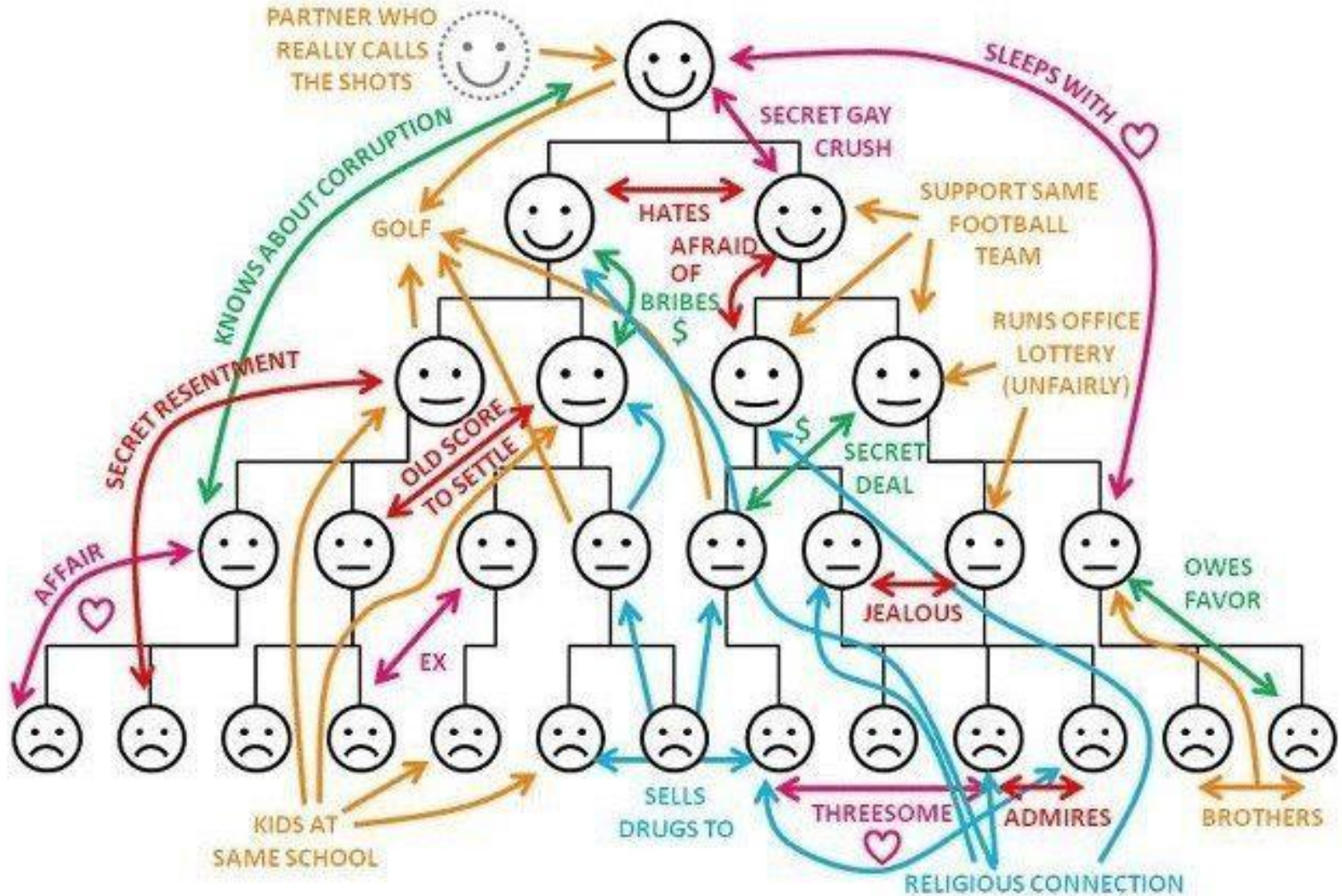


# IS AN ORGANIZATION A MACHINE?

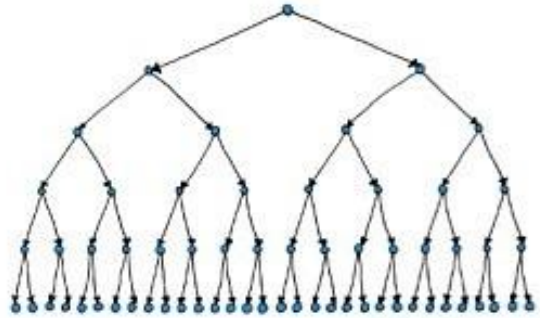
# WHAT KIND OF THE NETWORK OF AN ORGANISATION?



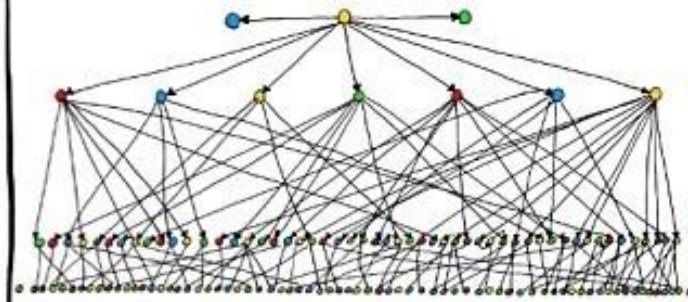
# The Real Organization Chart



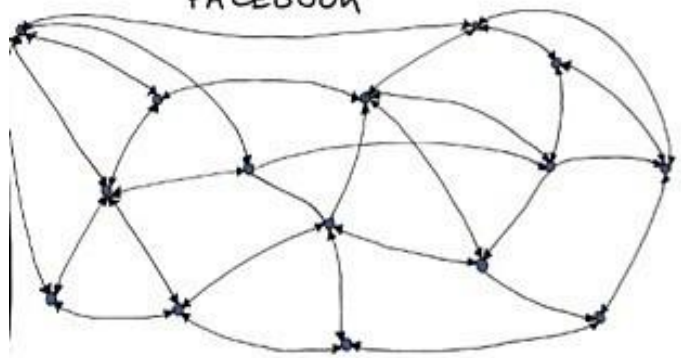
AMAZON



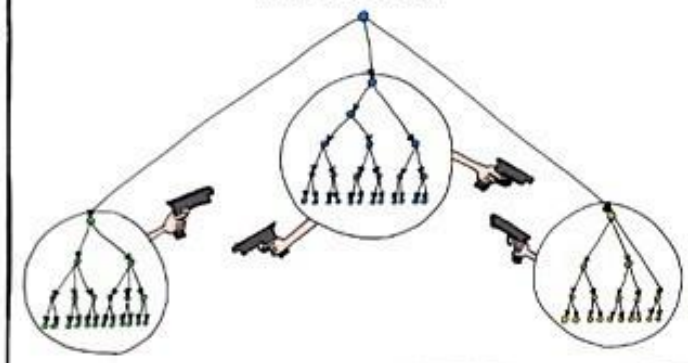
GOOGLE



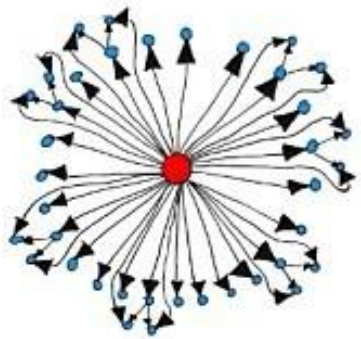
FACEBOOK



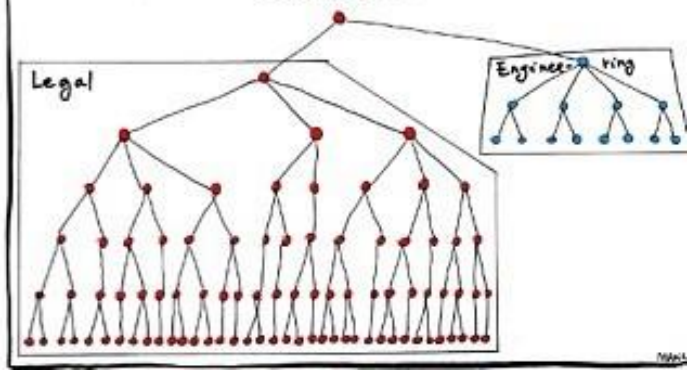
MICROSOFT



APPLE



ORACLE

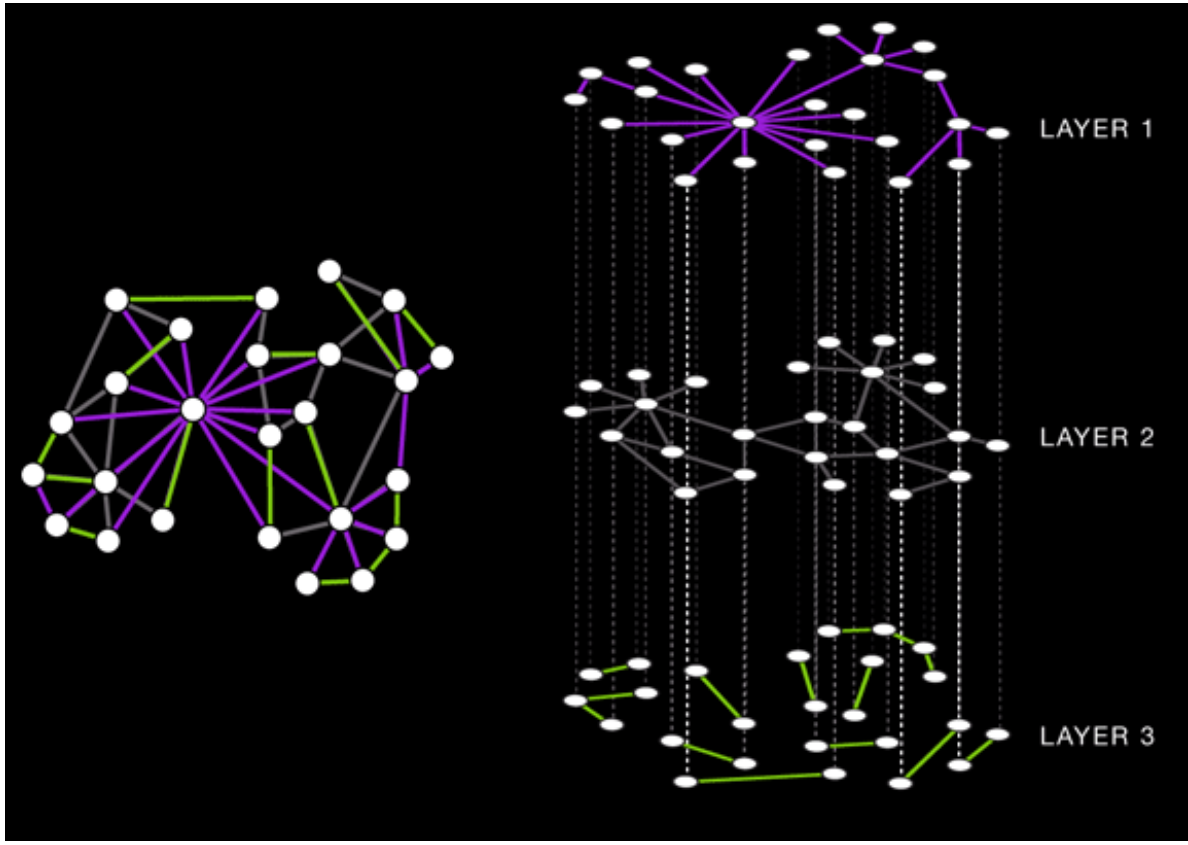



THE ORGANIZATION IS A NETWORK

# THERE ARE MORE THAN ONE NETWORKS

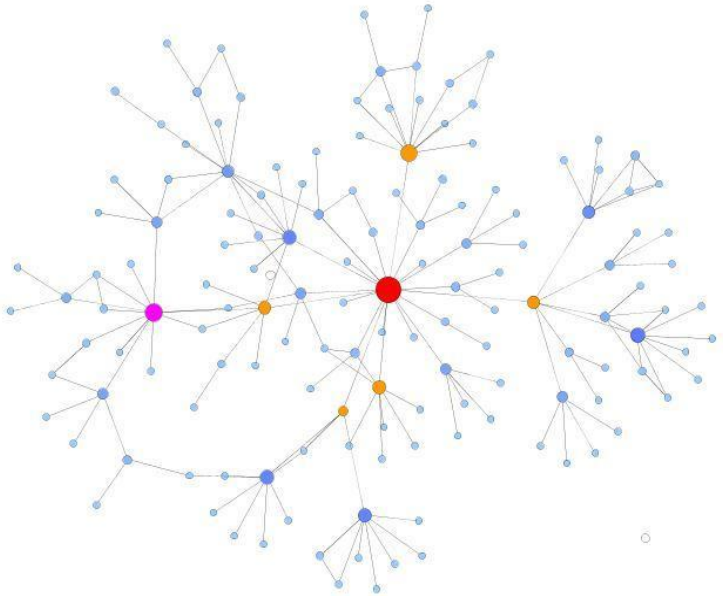
## DIFFERENT NETWORK LAYERS IN AN ORGANIZATION

- Personal power (from position)
- Professional ordering
- Knowledge sharing(s)
- Substitutional(s)
- Processes
- Spatial
- Personal preferences
- Gossip
- etc.

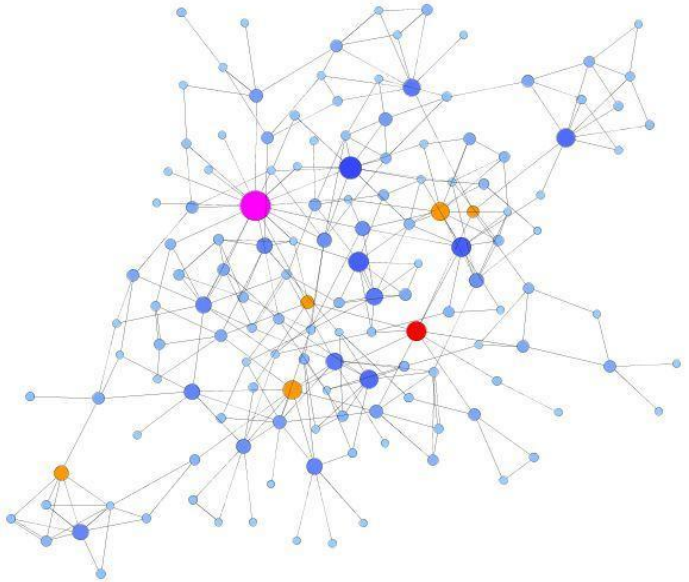


# REAULTS OF A CASE STUDY

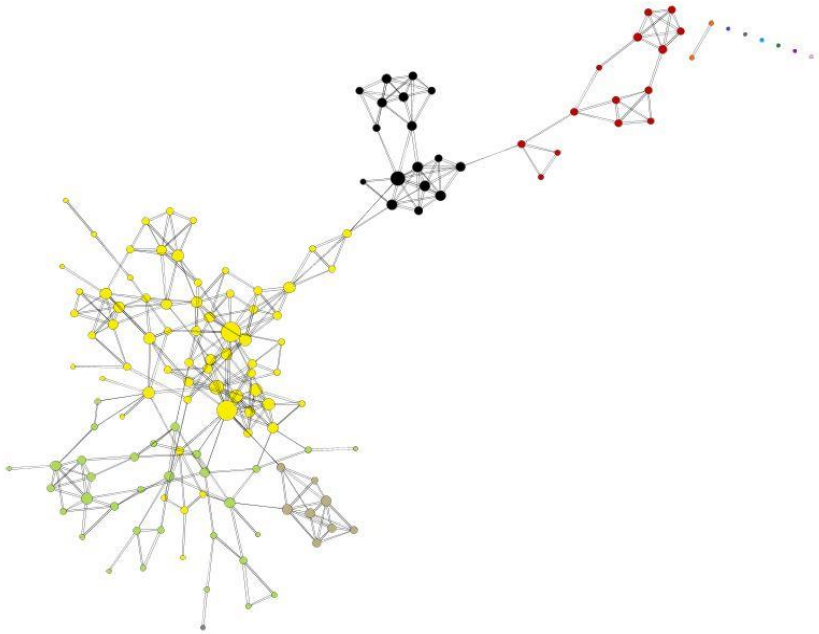
## KNOWLEDGE SHARING NETWORK LAYERS IN SAME ORGANIZATION.



Professional ordering



Voluntary knowledge sharing



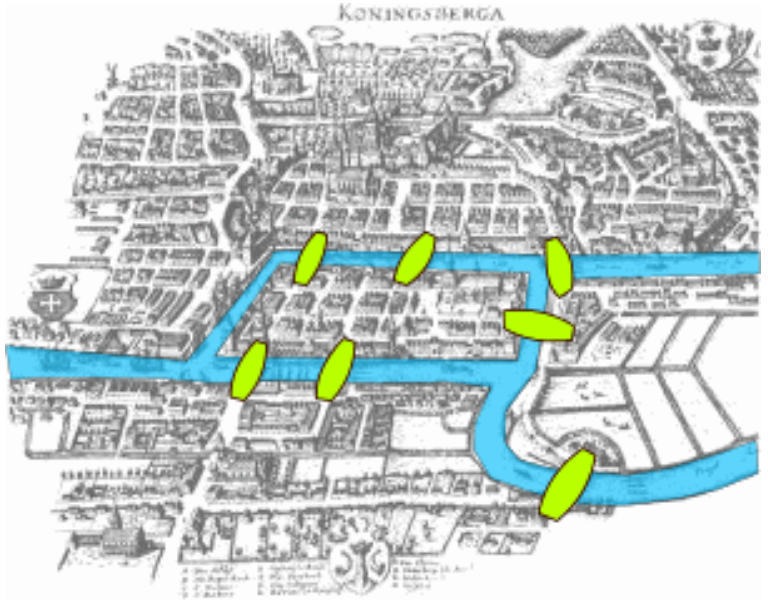
Gossip

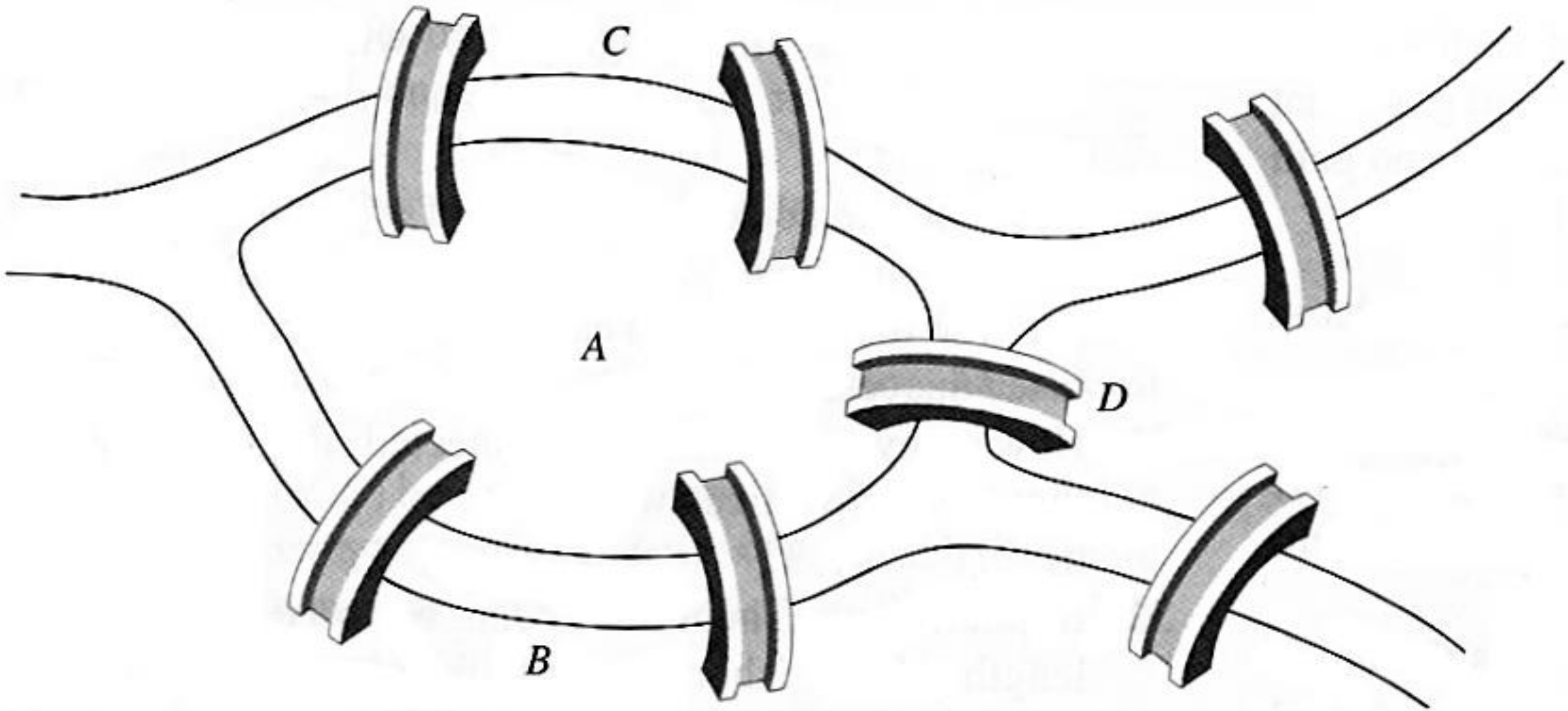
# HOW WAS THE GRAPH THEORY BORN ?

## THE PROBLEM OF BRIDGES OF KÖNIGSBERG

- Can we walk across all seven bridges and never cross the same twice?
- Leonhard Euler made a mathematical proof, that is not possible. (1741)

- It was the first graph,
- and the first time, that someone used a graph for solving a mathematical problem.

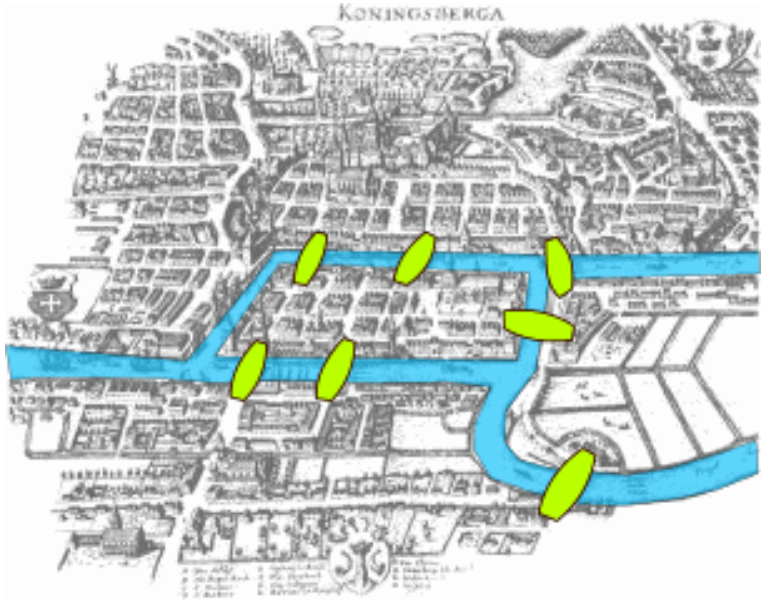



C
A
D
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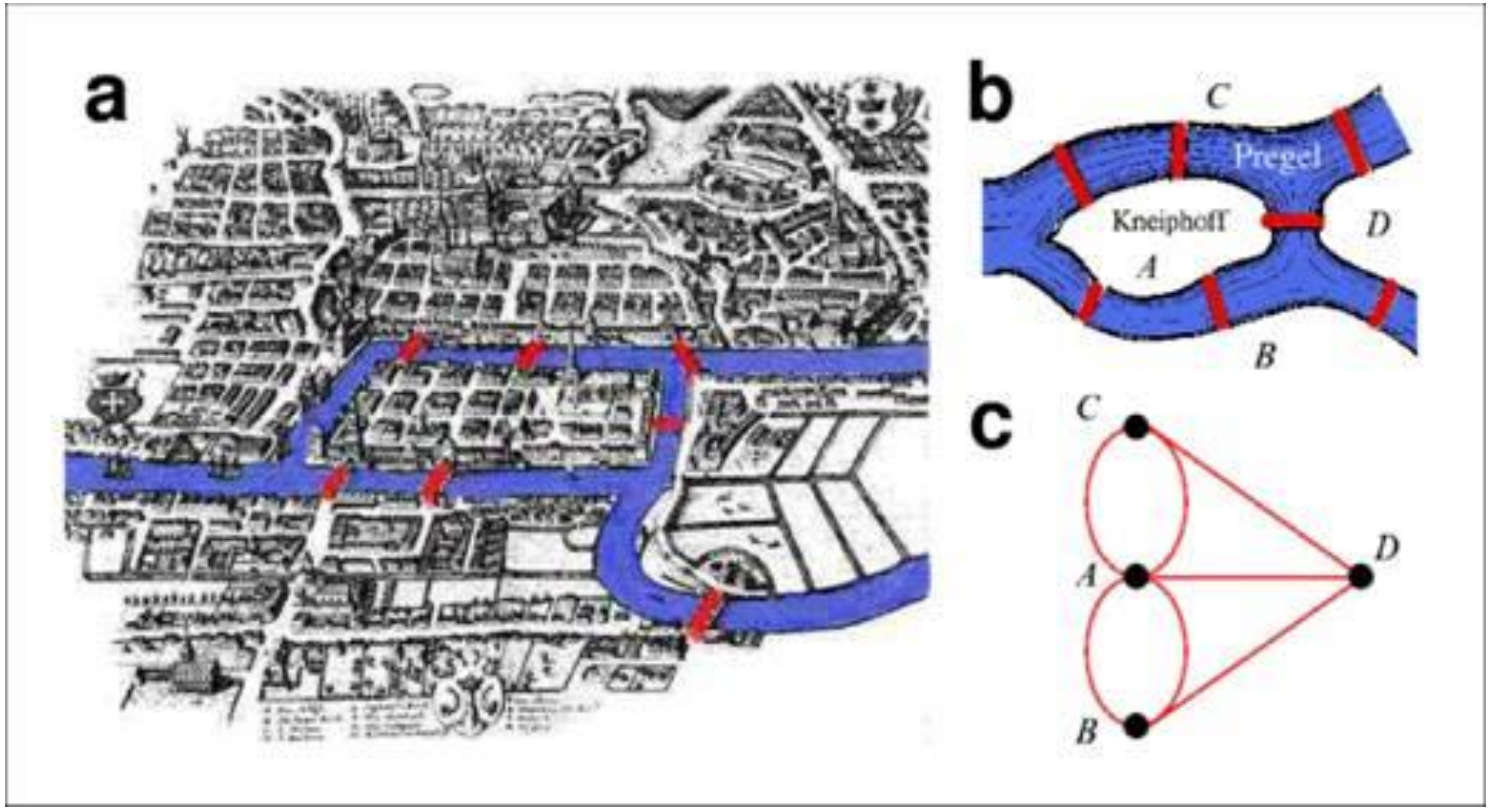
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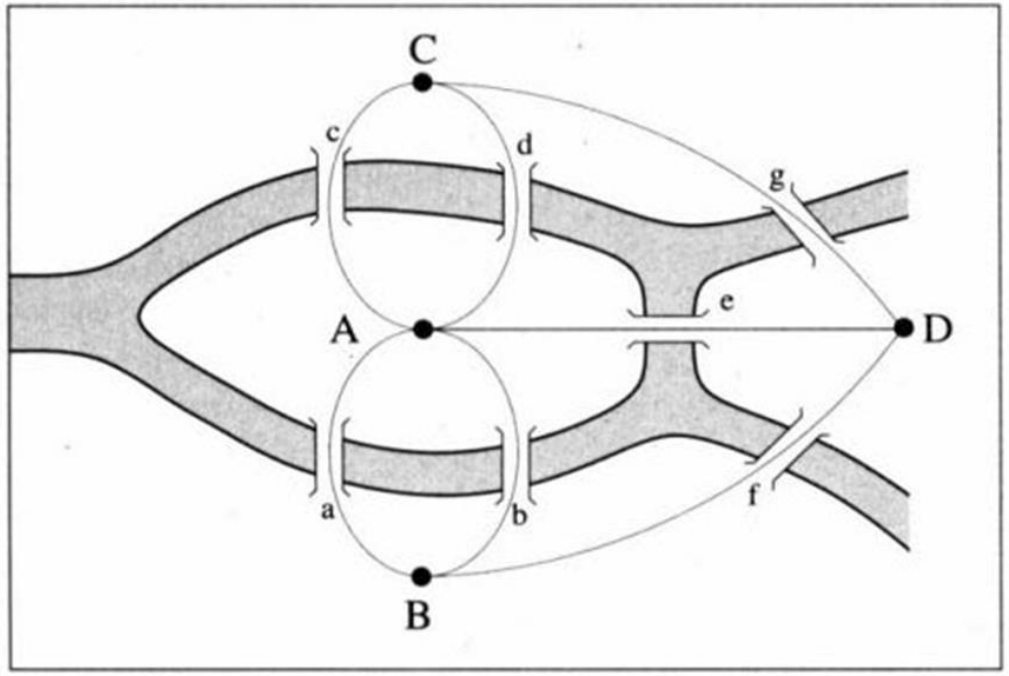
## THE PROBLEM OF BRIDGES OF KÖNIGSBERG

- Can we walk across all seven bridges and never cross the same twice?
- Leonhard Euler made a mathematical proof, that is not possible. (1741)
- He represented the four lands by letters.
- Connected those with lines, over the bridges.
- It was the first graph,
- and the first time, that someone used a graph for solving a mathematical problem.



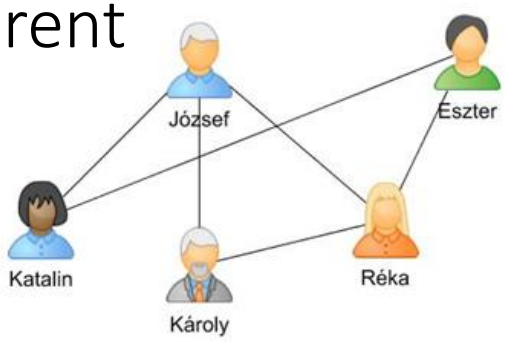
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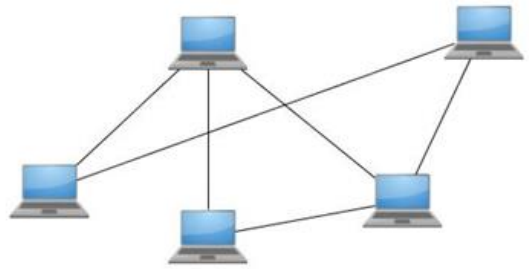


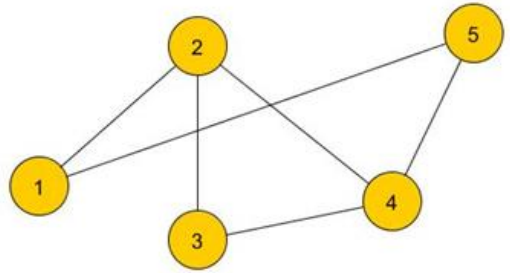
# NETWORKS AND GRAPHS

- We need to know how the components of the system interact with each other.
- We need a map of its wiring diagram – a graph.
- What is the different, between graphs and networks? – the reality.
- The graph theory explores how the different things are connected together but doesn't explore what these things are.
- The networks are graphs but real ones.

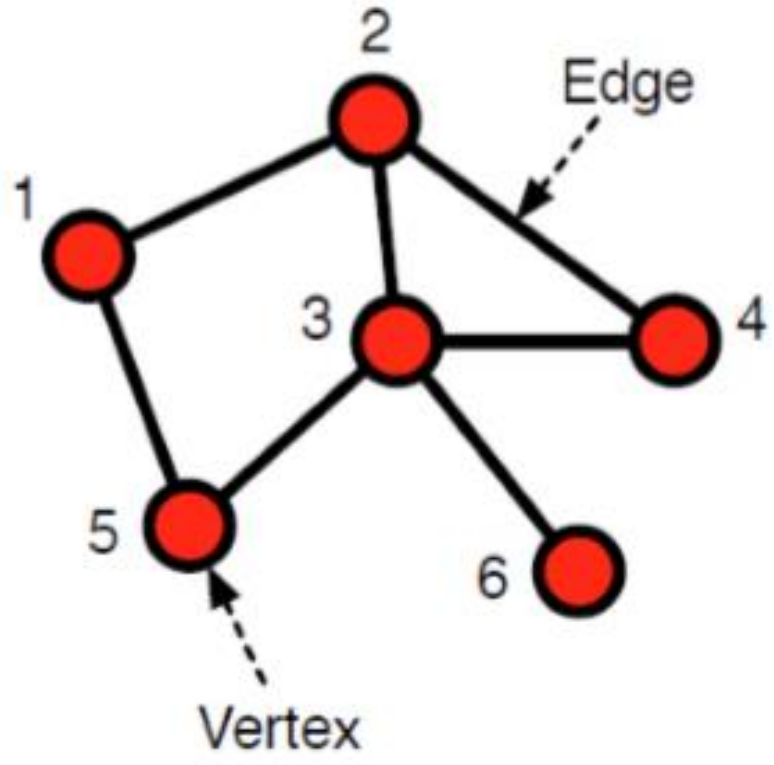


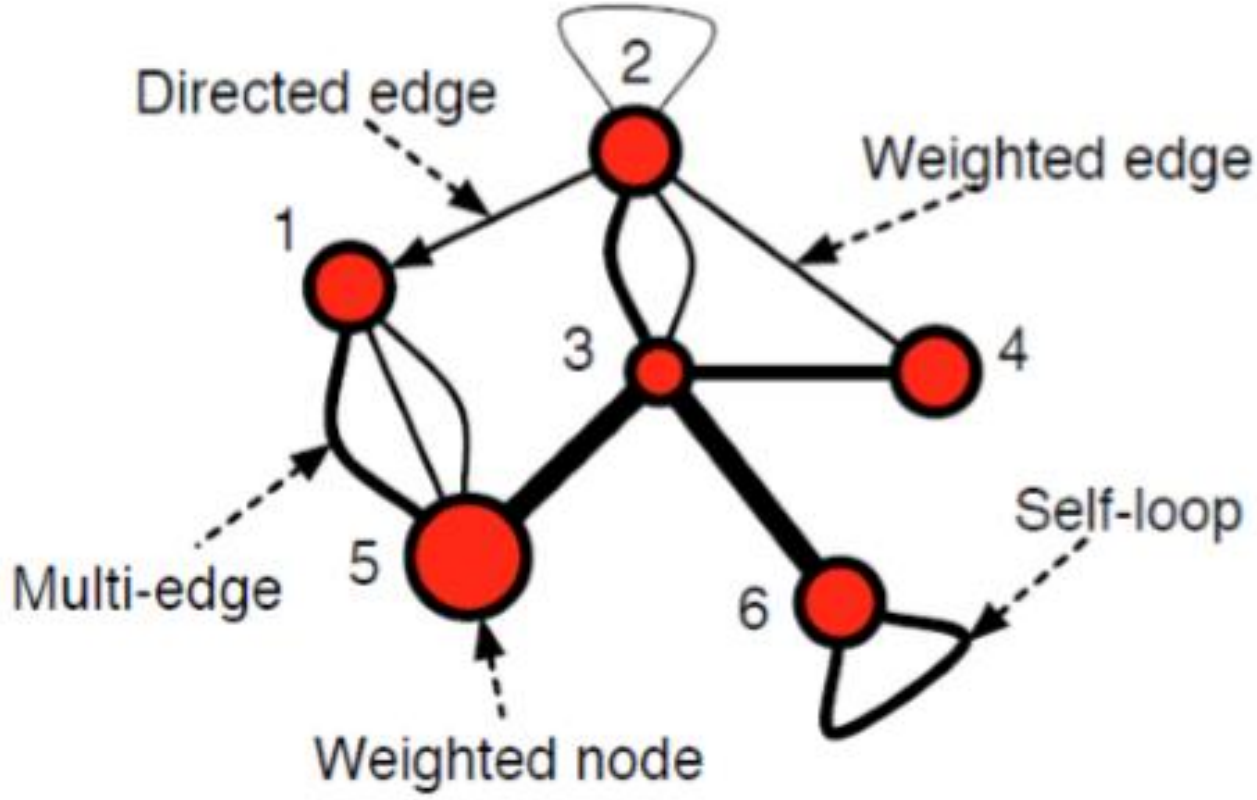



**The network science researches the connections of real things.**



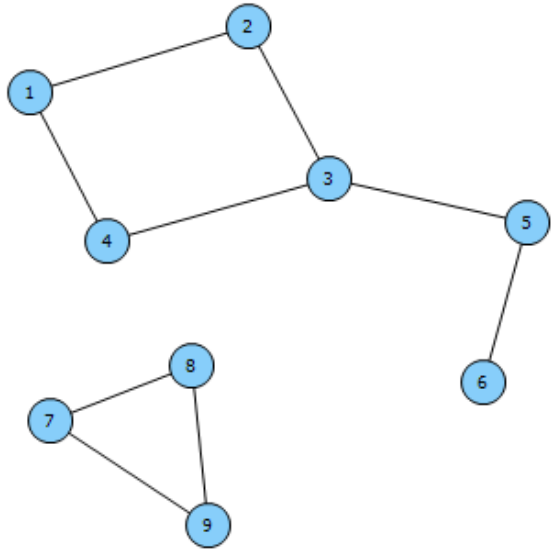
# THE PARTS OF GRAPHS

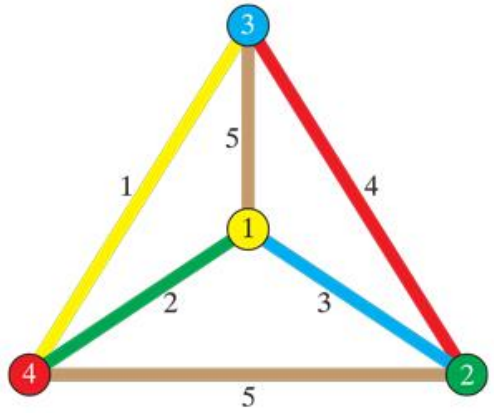


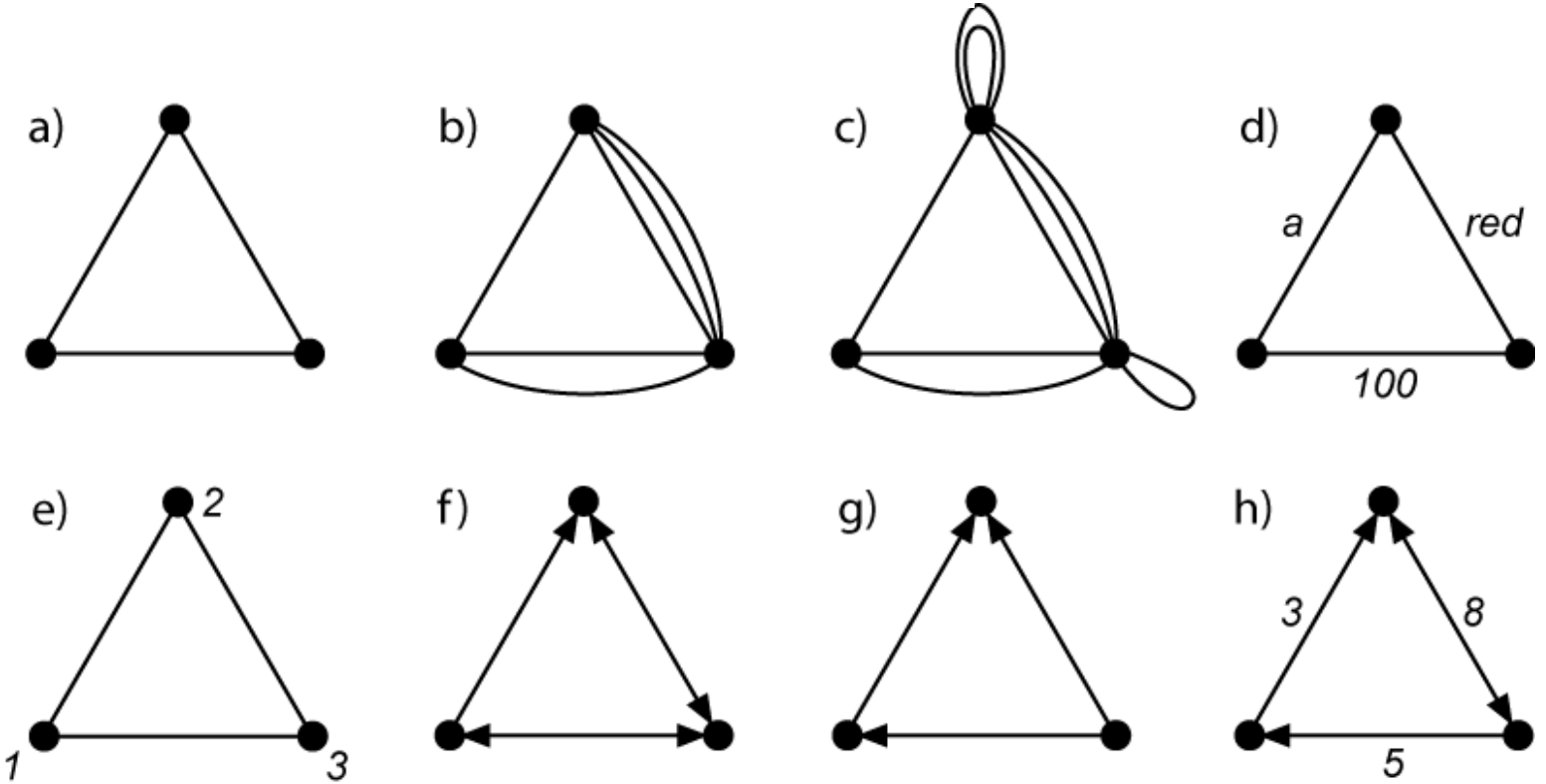



# THE TYPES OF GRAPHS

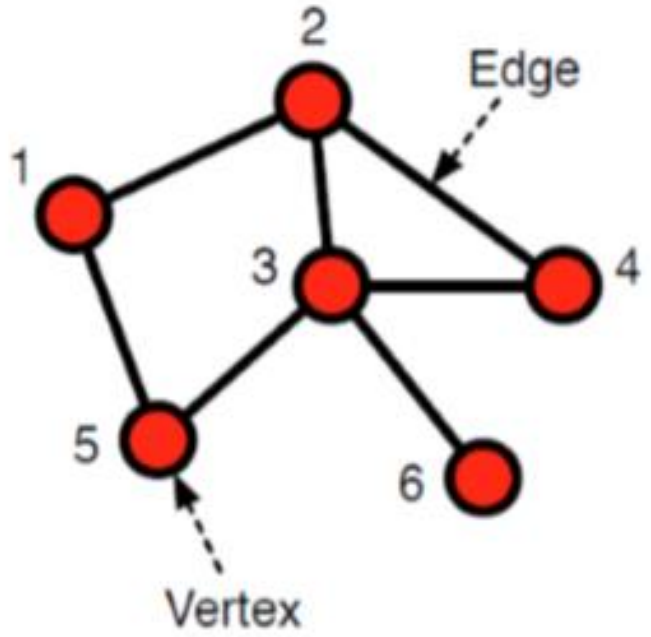
- **undirected/directed** graph
- **weighted/unweighted** graph
- **connected/disconnected** graph
- **simple/multi**graph
- **complete** graph
- **labelled** graph
- **oriented** graph
- **sub**graph



# THE GRAPH REPREZENTATIONS




**Adjacency matrix**

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}$$

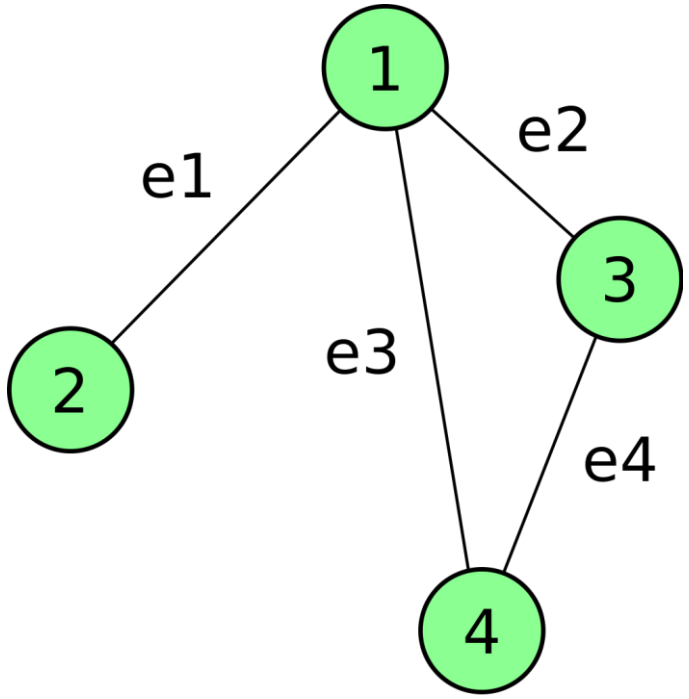
**Neighborhood list**

| | | | | |
|---|---|---|---|---|
| 1 | 2 | 5 | | |
| 2 | 3 | 1 | 4 | |
| 3 | 2 | 5 | 4 | 6 |
| 4 | 2 | 3 | | |
| 5 | 1 | 3 | | |
| 6 | 3 | | | |

$$\{(1,2),(1,5),(2,3),(2,4),(3,5),(3,6)\},$$

**Edge list**

# THE GRAPH REPREZENTATIONS

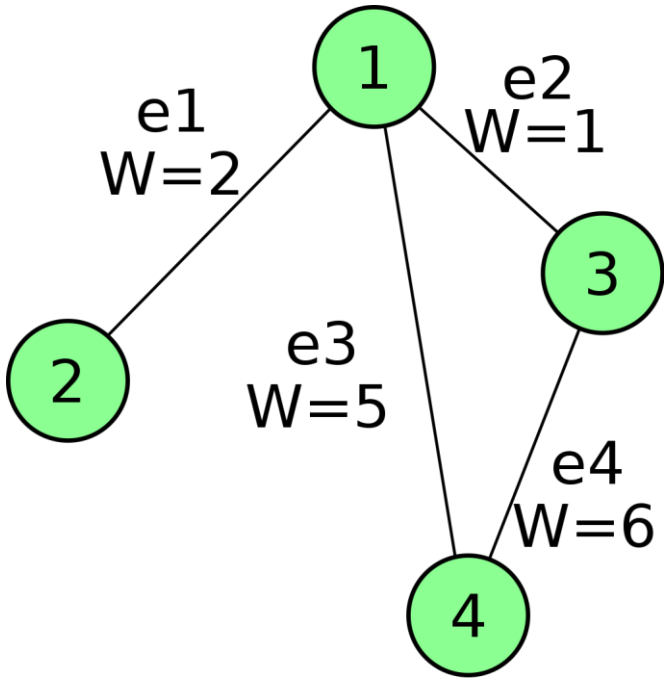



**Incidence matrix**

| | $e_1$ | $e_2$ | $e_3$ | $e_4$ |
|---|---|---|---|---|
| **1** | 1 | 1 | 1 | 0 |
| **2** | 1 | 0 | 0 | 0 |
| **3** | 0 | 1 | 0 | 1 |
| **4** | 0 | 0 | 1 | 1 |

$$= \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

# THE GRAPH REPREZENTATIONS




**Incidence matrix of weighted grapf**

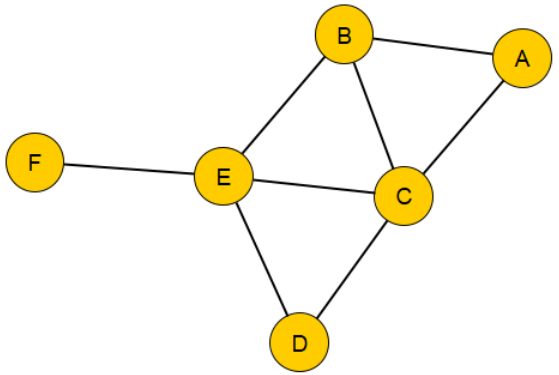
| | $e_1$ | $e_2$ | $e_3$ | $e_4$ |
|---|---|---|---|---|
| **1** | 2 | 1 | 5 | 0 |
| **2** | 2 | 0 | 0 | 0 |
| **3** | 0 | 1 | 0 | 6 |
| **4** | 0 | 0 | 5 | 6 |

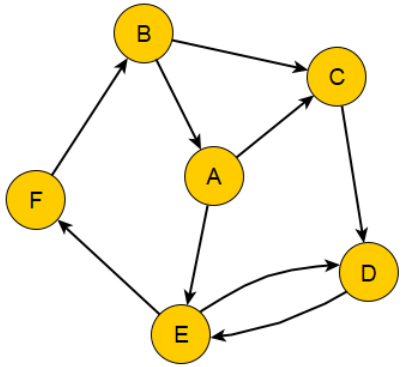
$$= \begin{bmatrix} 2 & 1 & 5 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 6 \\ 0 & 0 & 5 & 6 \end{bmatrix}$$

# EXERCISE

**Make the adjacency and incidence matrices of these graphs!**

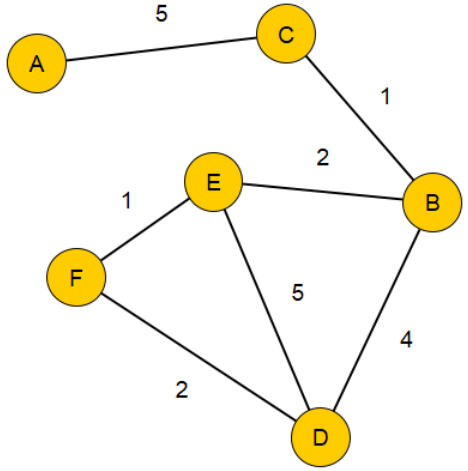
a.)

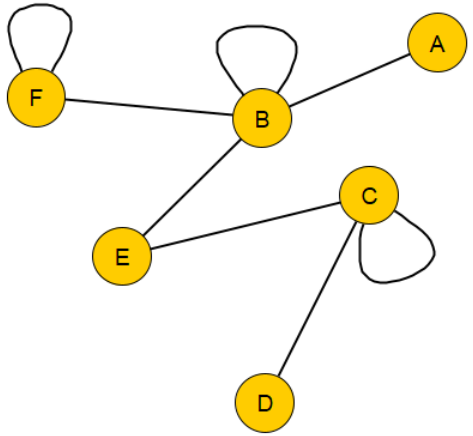



b.)

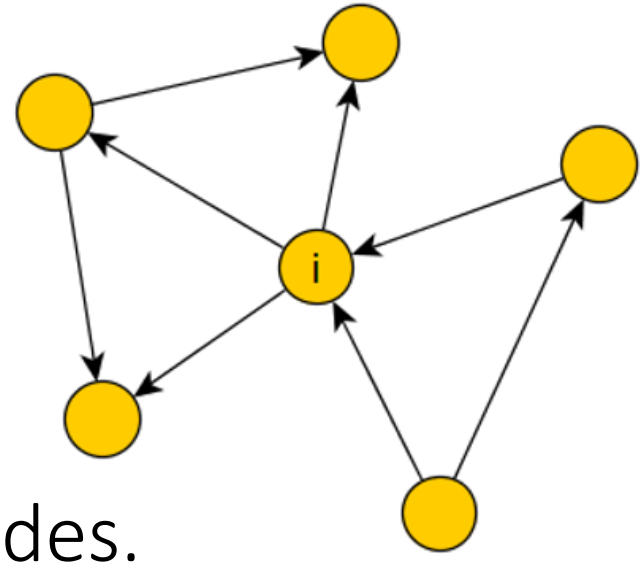
c.)




d.)

# GRAPH INDICATORS




## Degree

It represents the **number of links** it has to other nodes.

We denote with $k_i$ the degree of the node $i$ in the network.

- The most important property of the nodes.
- In a directed network there are indegree $k_i^{in}$ and outdegree $k_i^{out}$
- The indegree and the outdegree of a node could be different.

# GRAPH INDICATORS

## Average Degree

- In an undirected network:

$$\langle k \rangle = \frac{1}{N}\sum_{i=1}^{N} k_i = \frac{2L}{N}$$

- In a directed network:

$$\langle k^{in} \rangle = \frac{1}{N}\sum_{i=1}^{N} k_i^{in} = \langle k^{out} \rangle = \frac{1}{N}\sum_{i=1}^{N} k_i^{out} = \frac{L}{N}$$

# GRAPH INDICATORS

## Degree Distribution

- The degree distribution $p_k$ provides the probability that a random selected node in the network has degree $k$.
- Since $p_k$ is a probability it must be normalized.

$$\sum_{k=1}^{N} p_k = 1$$

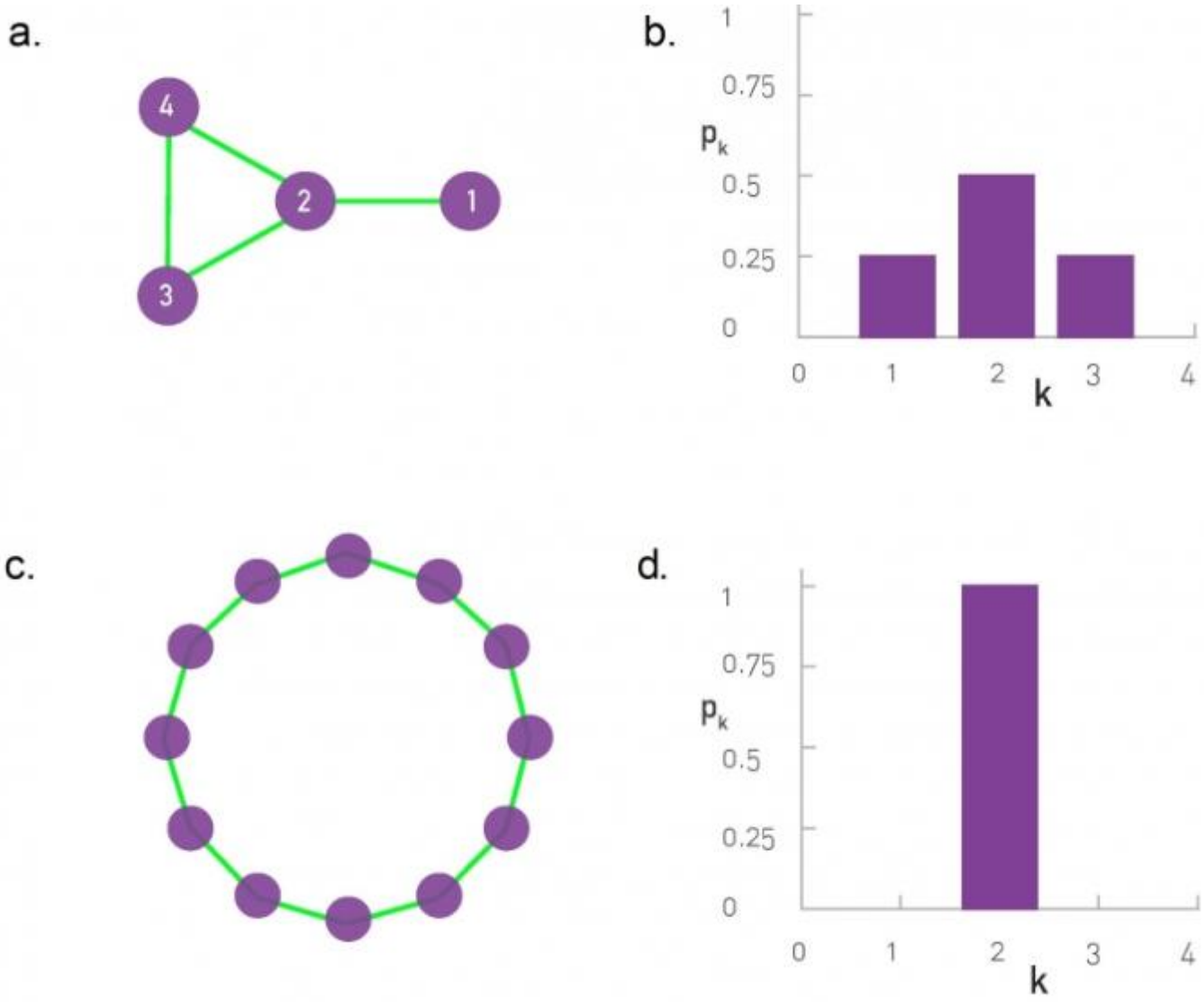
- For a network with $N$ nodes the degree distribution is the normalized histogram is given by

$$p_k = \frac{N_k}{N}$$

# GRAPH INDICATORS

## Degree Distribution

$$p_k = \frac{N_k}{N}$$

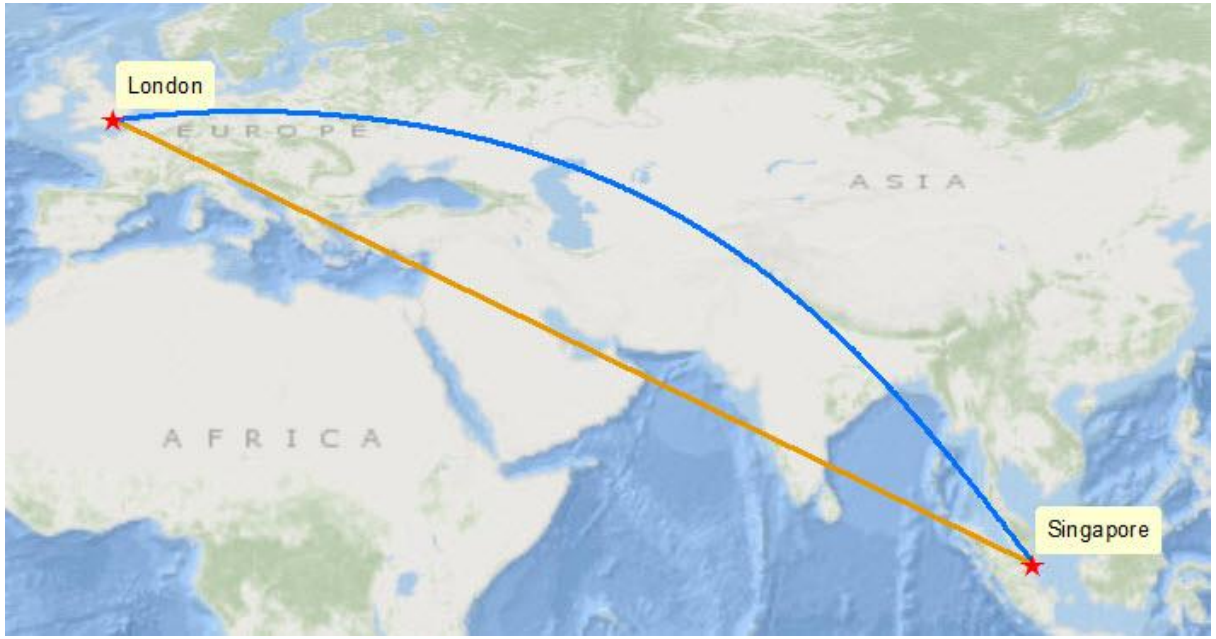



We use **column diagram** for visualization of it.

# DISTANCES IN THE NETWORKS

What does distance mean in a real networks?

- between two points?
- between two cities?
- between two railway stations?
- between two people?
- between two webpages?




**The physical distance is not always relevant in the networks.**

# DISTANCES IN THE NETWORKS

- We use the **path length** between two nodes.
- The path's length represents **the number of the links that the given path contains**.
- The network would be:
  - weighted or unweighted,
  - directed or undirected.

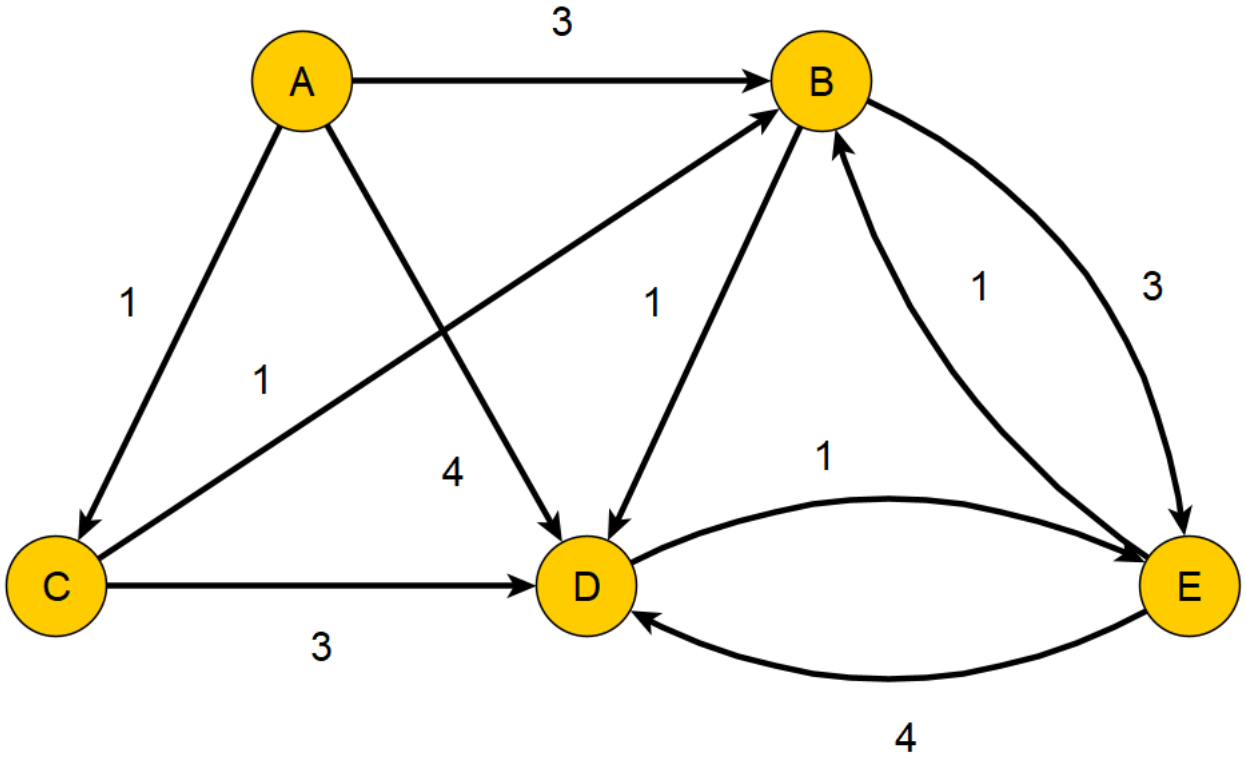
# DISTANCES IN THE NETWORKS

**Shortest path**

- The **shortest distance between two nodes**. (This is the path with the fewest number of links when the network is unweighted.)
- This is the distance $d_{ij}$ between node $i$ and $j$.
- In an undirected network $d_{ij} = d_{ji}$
- In a directed network would be $d_{ij} \neq d_{ji}$
- In a directed network the existence of a path from node $i$ to node $j$ does not guarantee the existence of a path from $j$ to $i$.

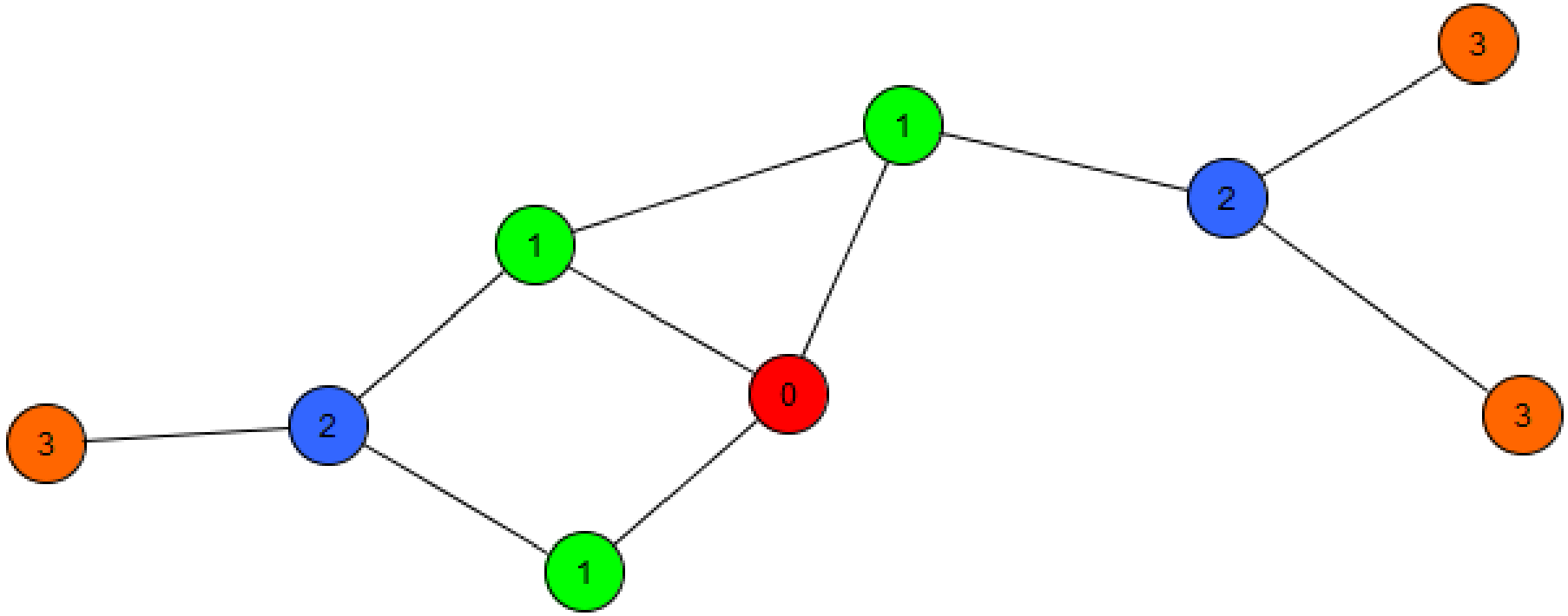
# DISTANCES IN THE NETWORKS

Finding the shortest path with **Dijkstra algorithm**



# DISTANCES IN THE NETWORKS

Finding the shortest path with **Breadth First Search (BFS) algorithm**



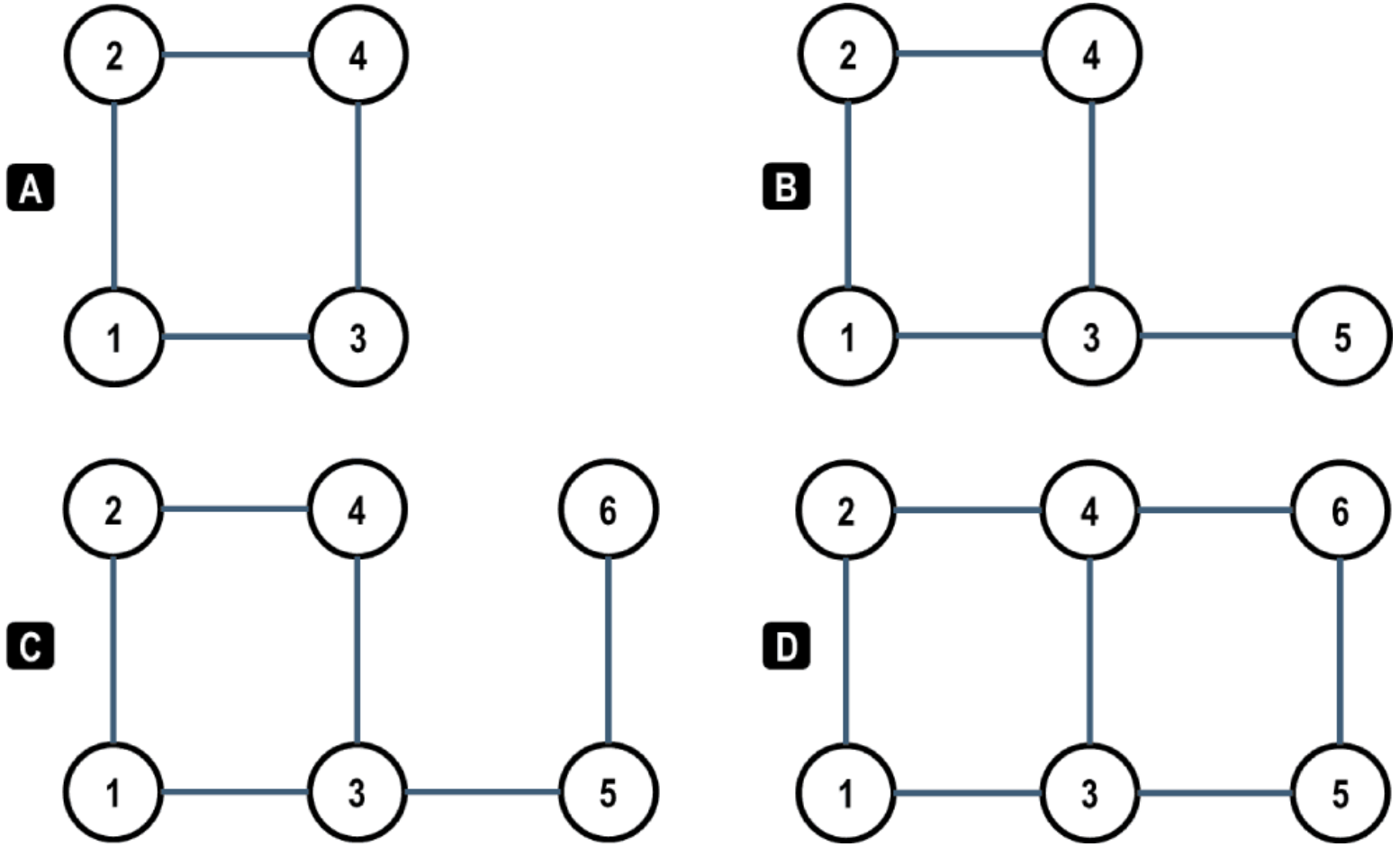
# DISTANCES IN THE NETWORKS

## Network diameter

- The network diameter $d_{max}$ is **the maximum of the shortest paths** in the network.

# DISTANCES IN THE NETWORKS

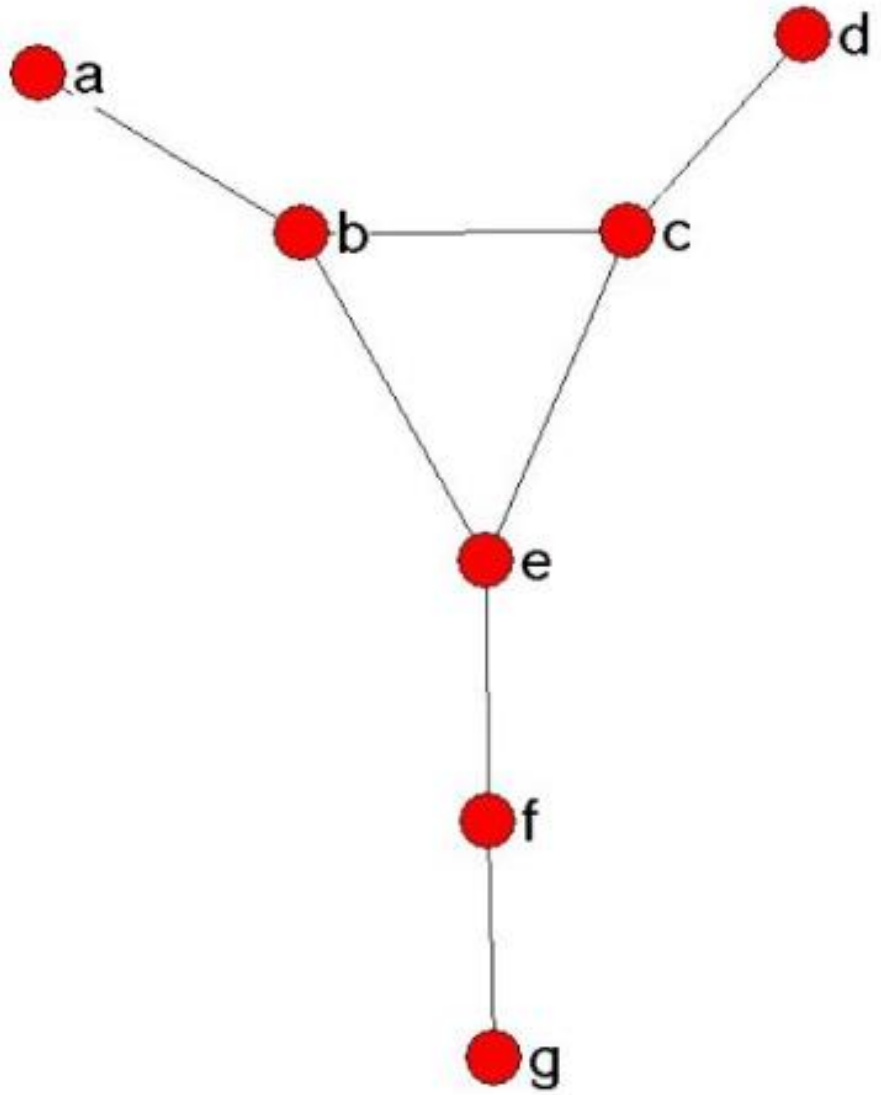
**How much is the network diameter's value?**



# DISTANCES IN THE NETWORKS

## The network distance matrix

| | a | b | c | d | e | f | g |
|---|---|---|---|---|---|---|---|
| a | 0 | 1 | 2 | 3 | 2 | 3 | 4 |
| b | 1 | 0 | 1 | 2 | 1 | 2 | 3 |
| c | 2 | 1 | 0 | 1 | 1 | 2 | 3 |
| d | 3 | 2 | 1 | 0 | 2 | 3 | 4 |
| e | 2 | 1 | 1 | 2 | 0 | 1 | 2 |
| f | 3 | 2 | 2 | 3 | 1 | 0 | 1 |
| g | 4 | 3 | 3 | 4 | 2 | 1 | 0 |



# DISTANCES IN THE NETWORKS

## Average Path Length

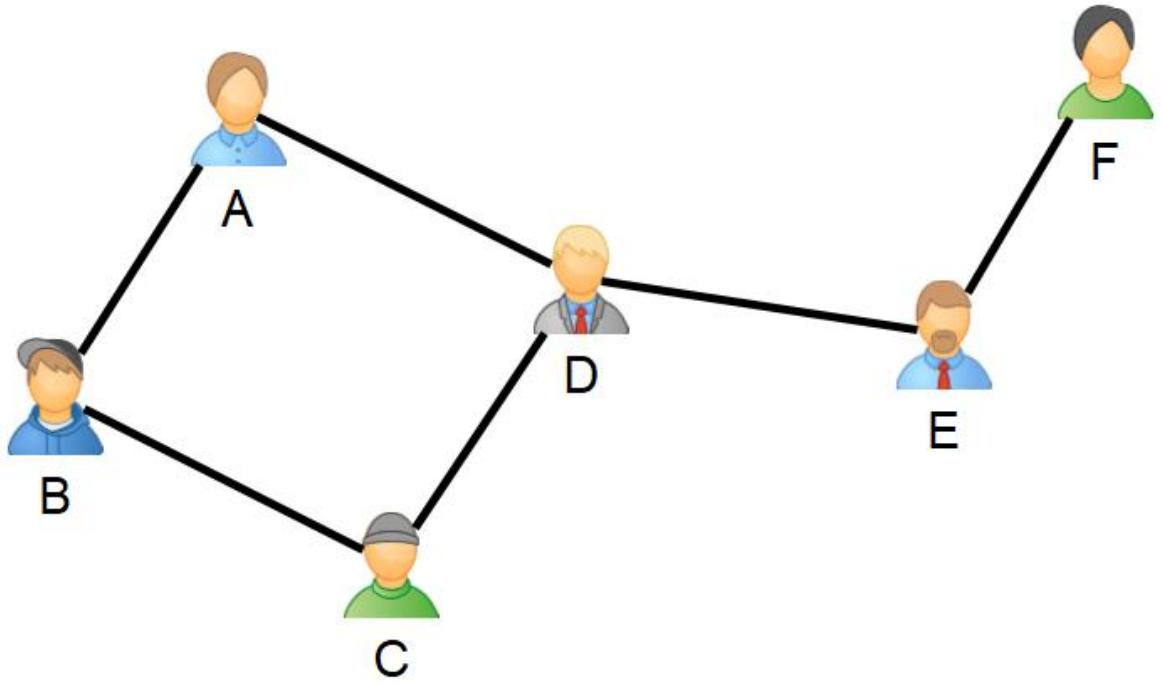
- The average path lenght $\langle d \rangle$ is the average shortest distance between all pairs of nodes in the network.
- In an undirected network $\langle d \rangle$ is

$$\langle d \rangle = \frac{1}{N(N-1)} \sum_{i,j=1;i\neq j}^{N} d_{ij}$$

# DISTANCES IN THE NETWORKS

1. Make the network distance matrix!
2. Calculate the average path length!

$$\langle d \rangle = \frac{1}{N(N-1)} \sum_{\substack{i,j=1 \\ i \neq j}}^{N} d_{ij}$$




$$\begin{bmatrix} 0 & 1 & 2 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 & 3 & 4 \\ 2 & 1 & 0 & 1 & 2 & 3 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 2 & 3 & 2 & 1 & 0 & 1 \\ 3 & 4 & 3 & 2 & 1 & 0 \end{bmatrix}$$

# BETWEENNES CENTRALITY

It shows how many shortest path go trough on a given node from the total shortest path of network.

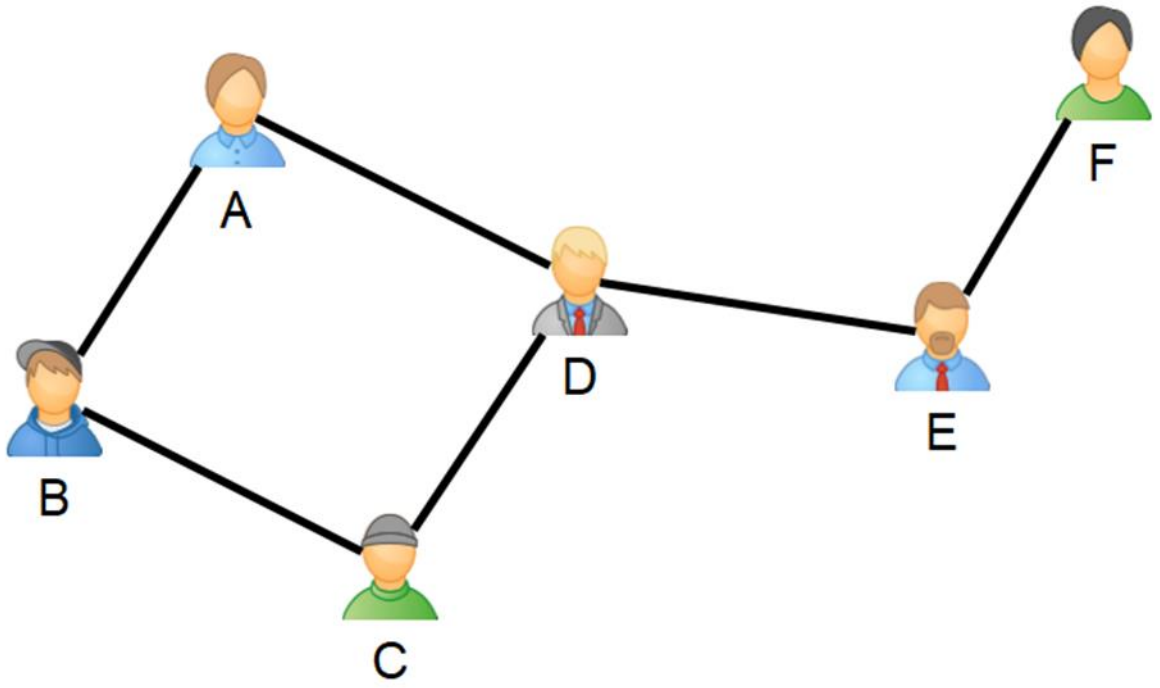
In undirected and unweighted networks:

$$C_b(n) = \sum_{\substack{i,j,n=1 \\ i \neq j \neq n}}^{N} \frac{g_{ij}(n)}{g_{ij}}$$

# BETWEENNES CENTRALITY

$$C_b(n) = \sum_{\substack{i,j,n=1 \\ i \neq j \neq n}}^{N} \frac{g_{ij}(n)}{g_{ij}}$$

Calculate the betweennes centrality of D!

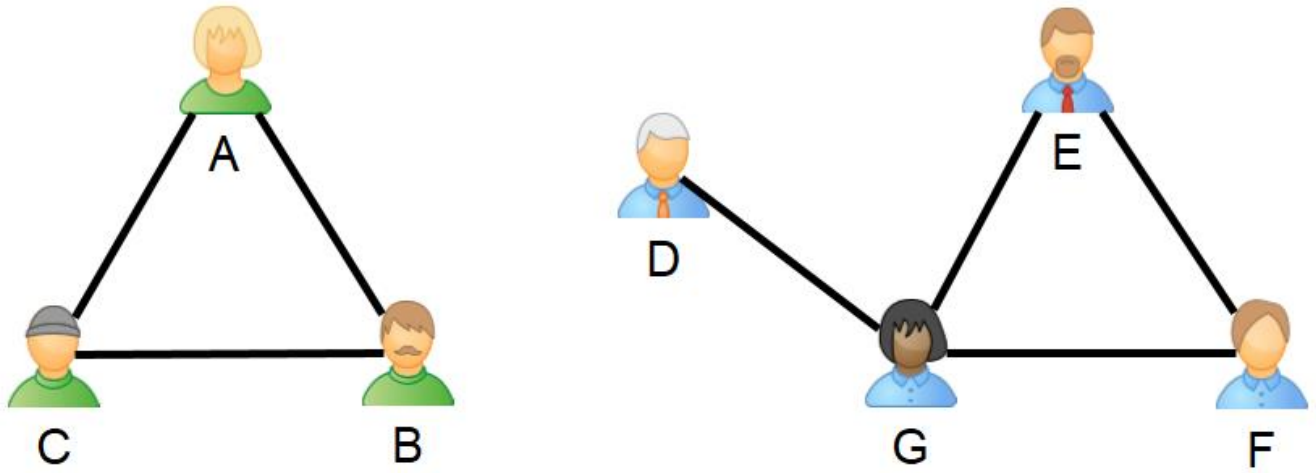


# CONNECTEDNESS

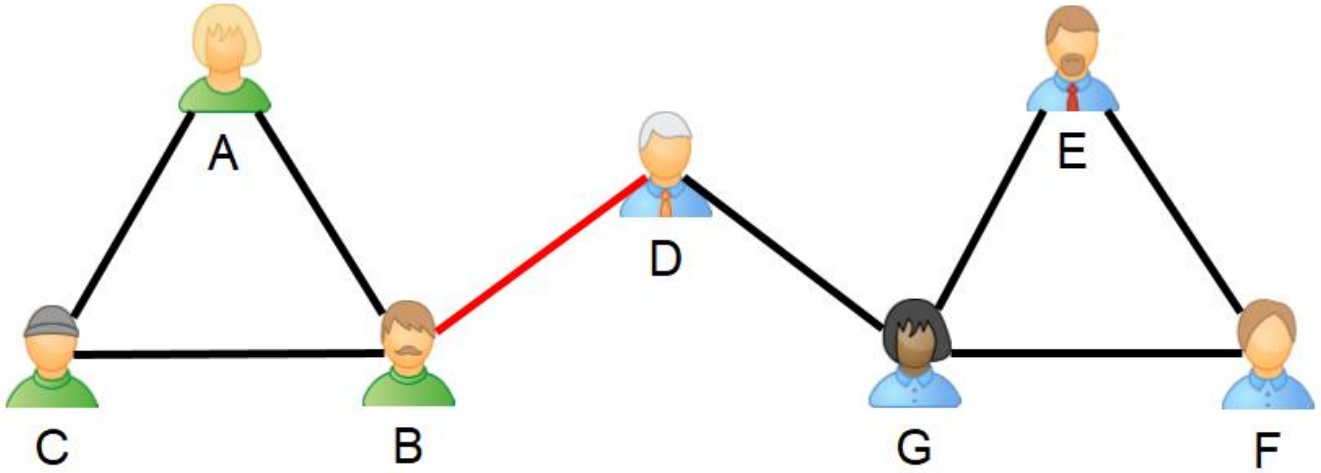
The key utility of most networks: **they ensure connectedness**.

- In an undirected network two nodes are **connected** if there is a path between them.
- They are **disconnected** if there isn't path between them. $d_{ij} = \infty$
- A **network is connected** if all pairs of nodes in the network are connected.
- A **network is disconnected** if there is at least one pair with $d_{ij} = \infty$
- In this case there are two or more **components** or **clusters**.

# CONNECTEDNESS




$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$




$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

# CLASTERING COEFFICIENT

The (local) **clustering coefficient** captures the degree to which the neighbors of a given node link to each other.

For a node $i$ with degree $k_i$ the local clustering coefficient is defined as:

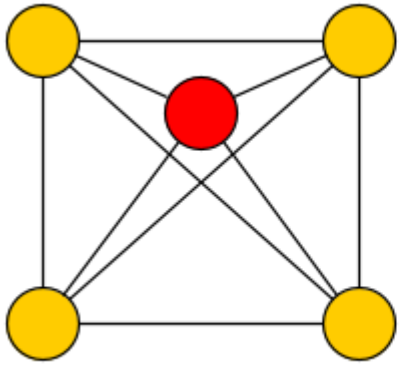
$$C_i = \frac{2L_i}{k_i(k_i - 1)}$$

where $L_i$ represents the number of links between the $k_i$ neighbors of node $i$. Note that $C_i$ is between 0 and 1.

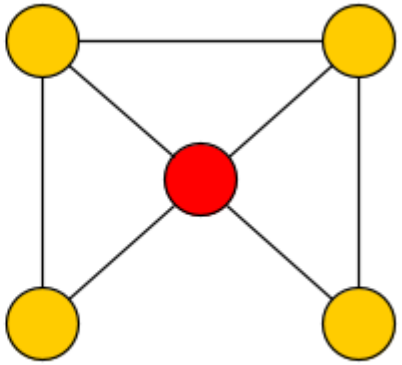
# CLASTERING COEFFICIENT

$$C_i = \frac{2L_i}{k_i(k_i - 1)}$$

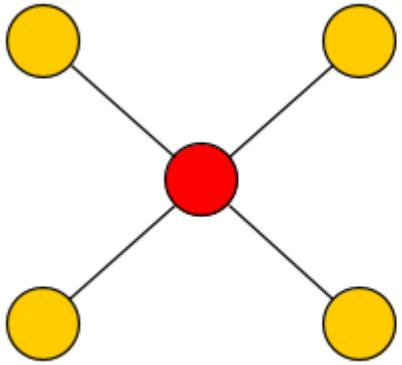
How much the clustering coefficient?



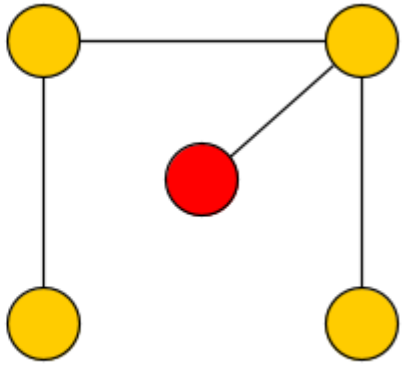
$C_i = 1$



$C_i = 1/2$



$C_i = 0$



$C_i = 0$

# CLASTERING COEFFICIENT

The degree of clustering of a **whole network** is captured by the **average clustering coefficient** $\langle C \rangle$, representing the average of $C_i$ over all nodes $i = 1, \dots, N$.

In undirected and unweighted networks:

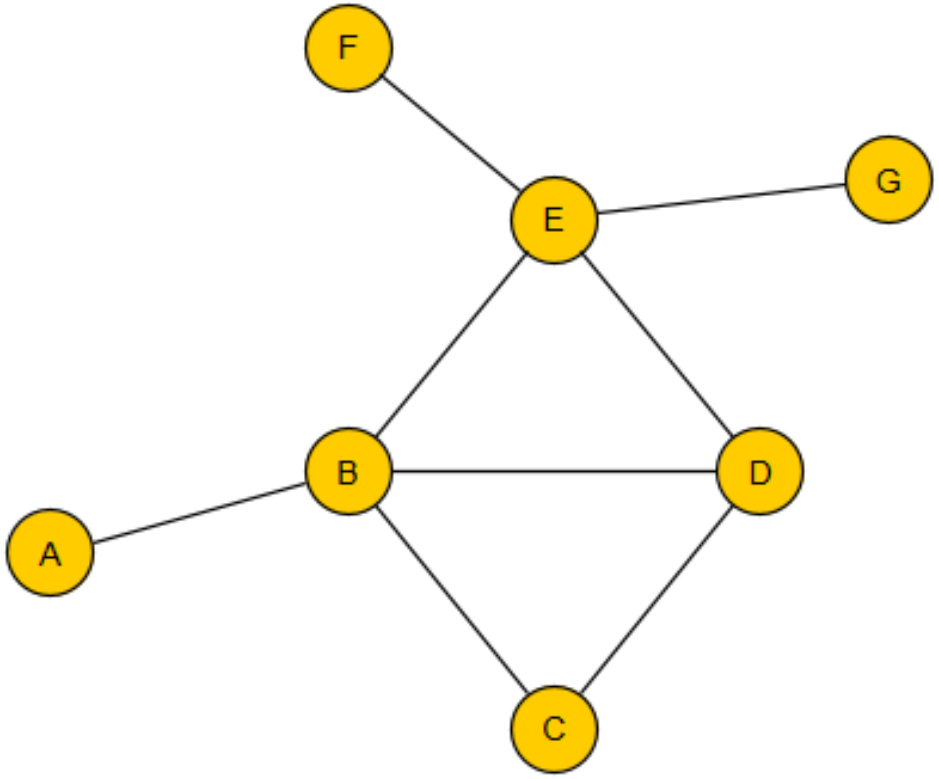
$$\langle C \rangle = \frac{1}{N} \sum_{i=1}^{N} C_i$$

# CLASTERING COEFFICIENT

$$C_i = \frac{2L_i}{k_i(k_i - 1)} \qquad \langle C \rangle = \frac{1}{N}\sum_{i=1}^{N} C_i$$

**Calculate**:

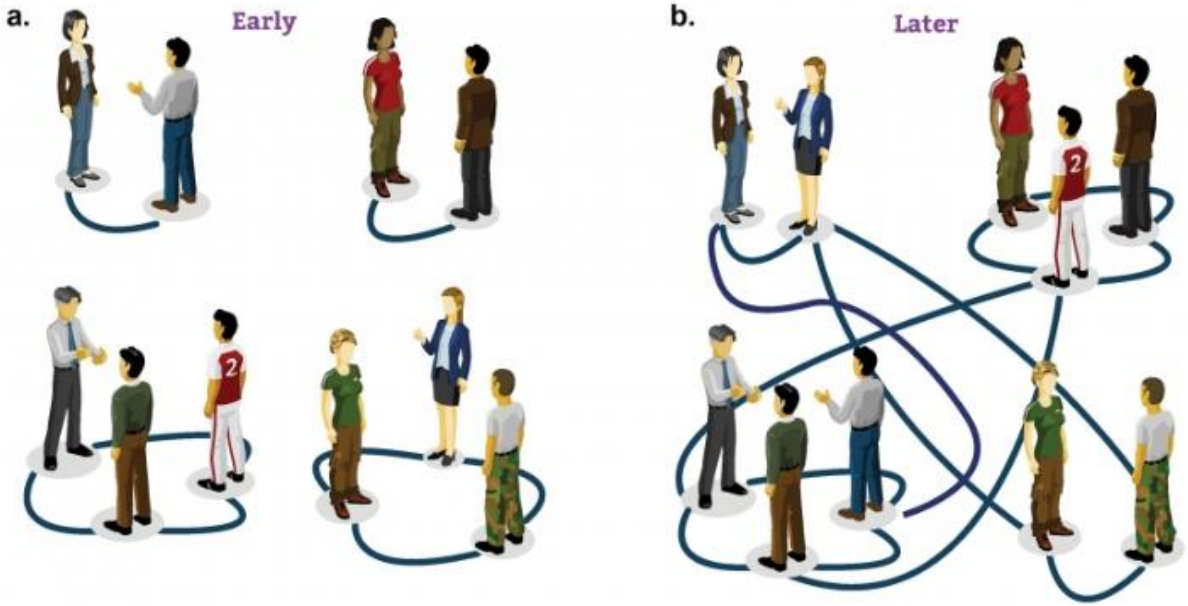
- the local clastering coefficient for each nodes,
- the average clastering coefficient!



# TYPES OF NETWORKS

**Let’s have a party!**

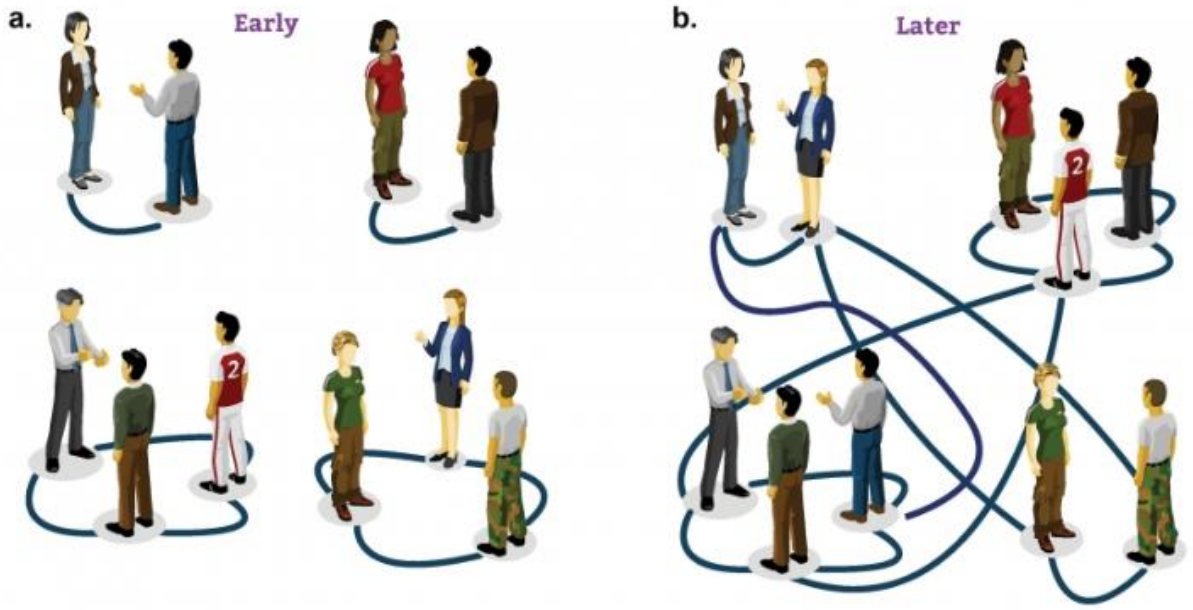
- Take 100 people who initially do not know each other.
- They start chatting in two or three person groups.
- But the guests will be bored when they talk with same person too long time.
- They look for new persons to talking.



# TYPES OF NETWORKS

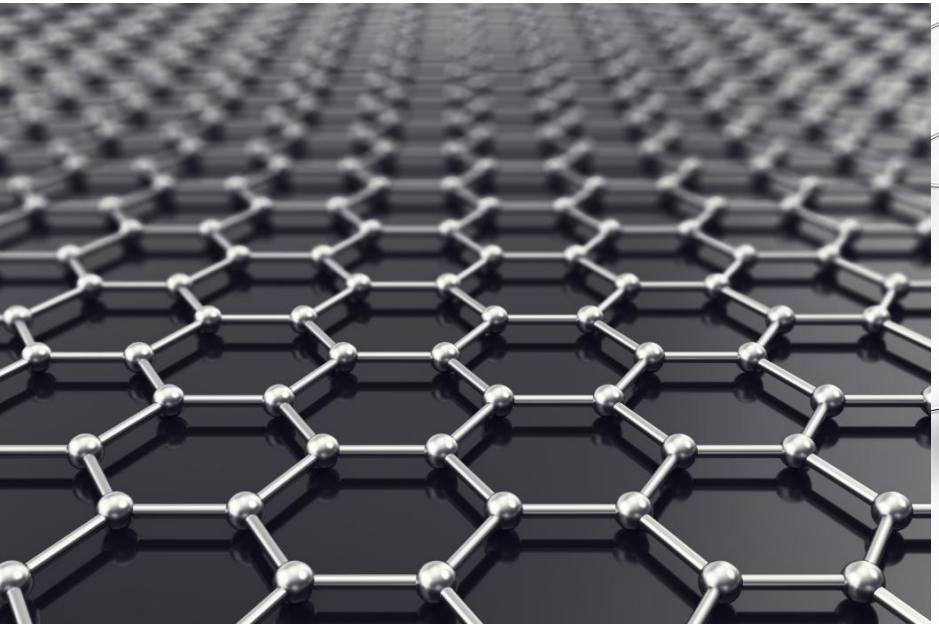
**Let’s have a party!**

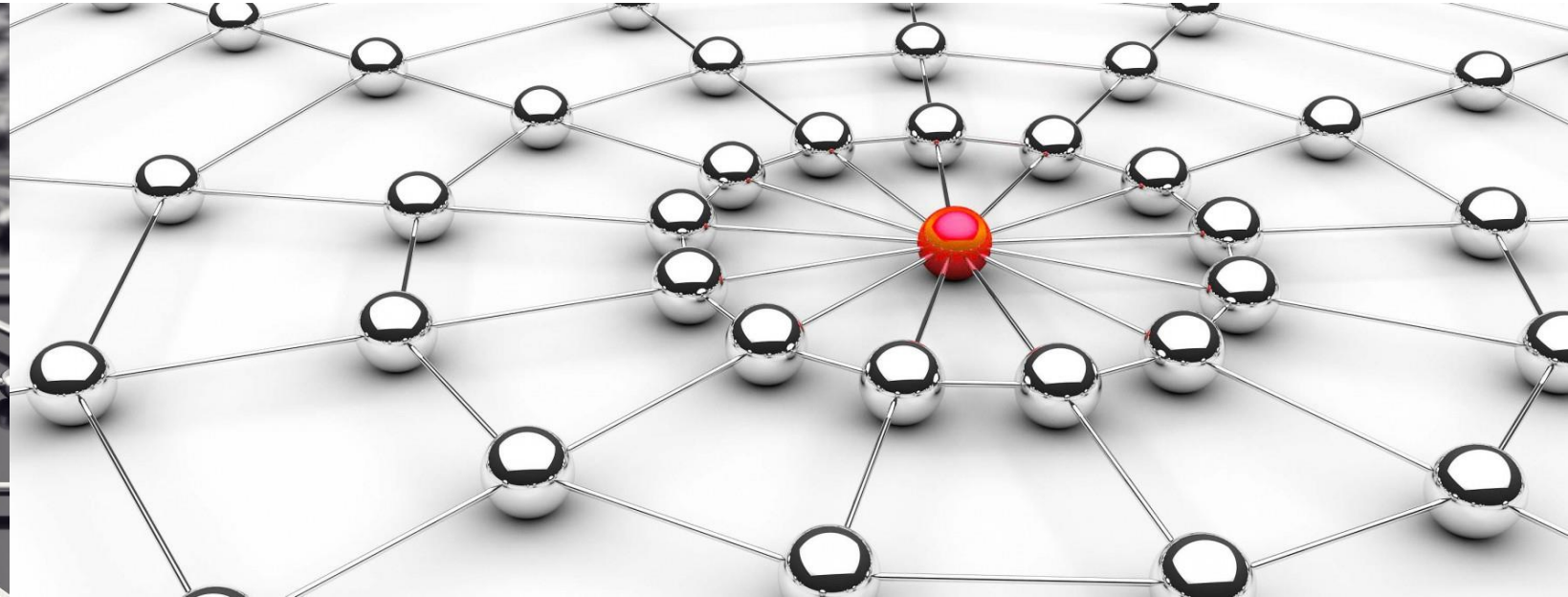
- Start an information by wine for one person only.
- Hence in no time everyone will know this „secret” information.
- This information will leak into the group of the guests slowly.
- Let’s calculate!
- How many time has the red labeled wine?



# TYPES OF NETWORKS

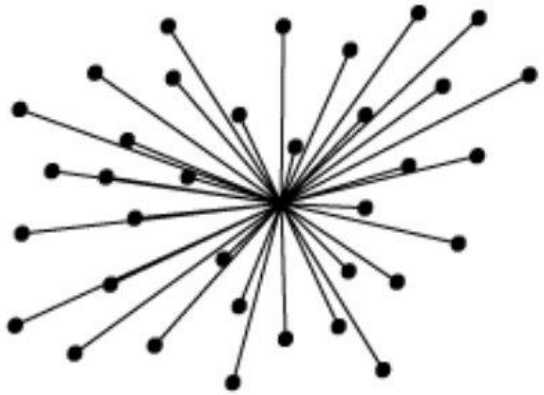
- Which network models reproduce the properties of real networks?
- Are the real networks similar that the crystal lattices?
- Are the real networks similar that the spider webs?



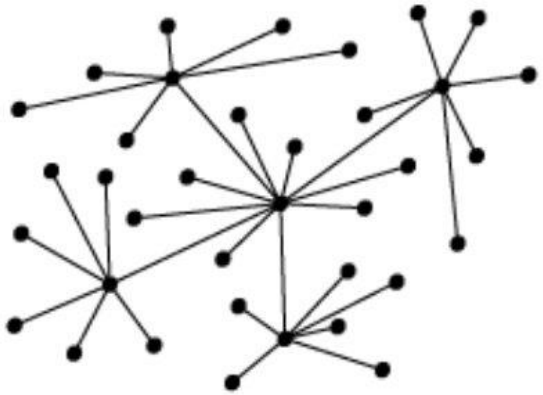


# TYPES OF NETWORKS

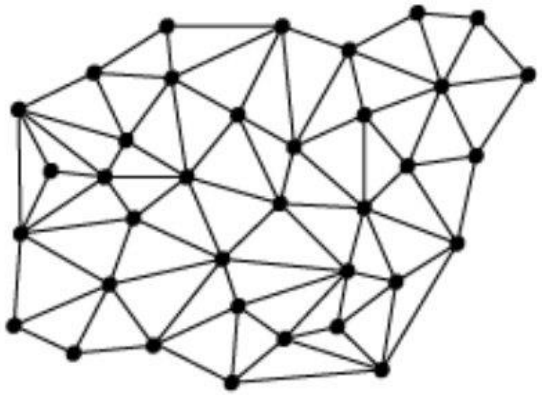
- The basic parts of every network are same: nodes and links.
- But the connection structure (topology) would be different!
- The properties of the network depend on topology.
- The Paul Baran type topologies:



centralised



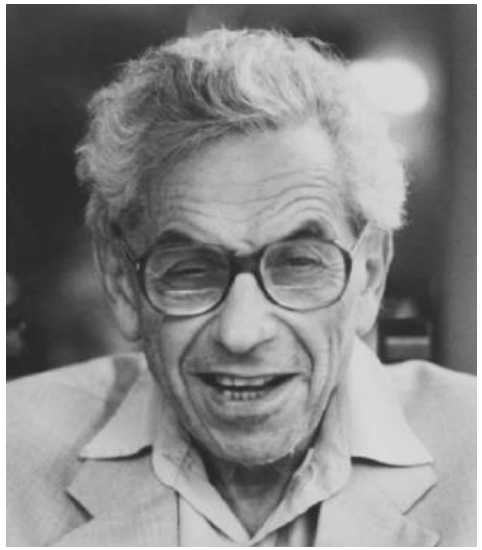
decentralised



distributed

# RANDOM NETWORKS

- Anatol **Rapoport** and Ray **Solomonof** discovered that:
  - if increase the average degree in a network than appears a connected **giant component**.
- Pál **Erdős** and Alfréd **Rényi** merged the probability theory and combinatorics with graph theory.
- The random networks often are called **Erdős-Rényi (ER) networks**.
- Edgar Nelson **Gilbert** introduced the random networks indepently too.





# RANDOM NETWORKS

**$G(N, p)$ model: Each pair of $N$ labeled nodes is connected with probability $p$.**

Let's make **random network**:

1. Make $N$ isolated nodes.
2. Select a node pair and generate a $p$ random number between 0 and 1.
3. If the number exceeds the $p$ then connect the nodes, otherwise leave disconnected them.
4. Repeat the step 2. for each of the $N(N-1)/2$ node pairs.

# RANDOM NETWORKS

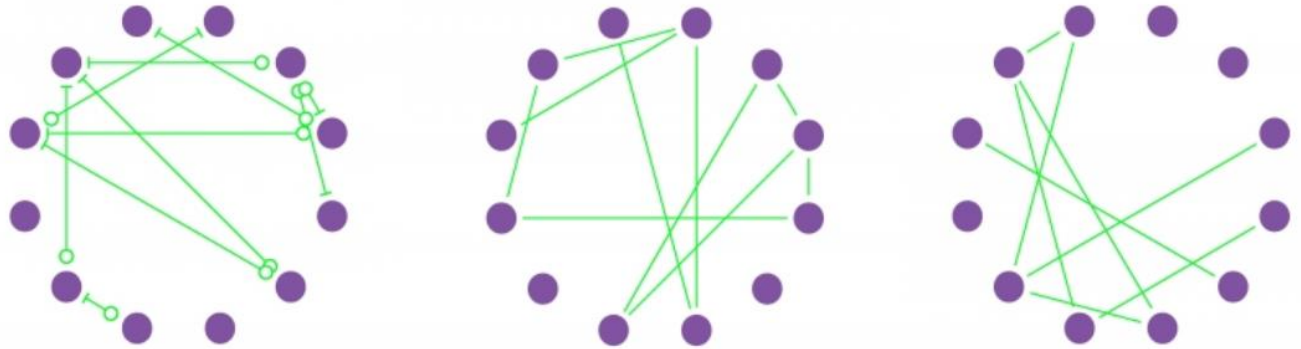
**The average degree of random network**

$$\langle k \rangle = \frac{2\langle L \rangle}{N} = p(N-1)$$

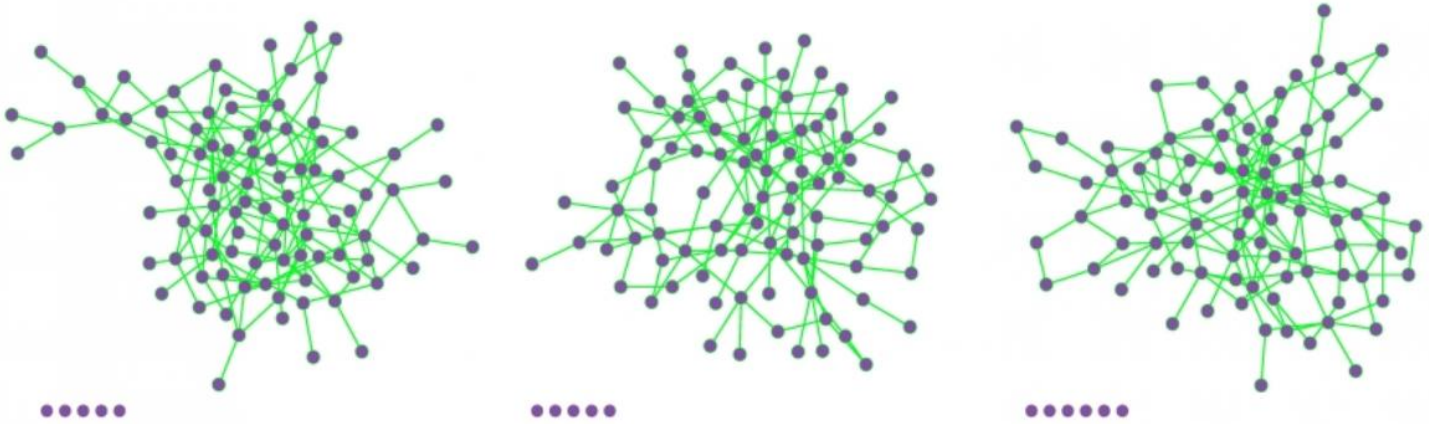
# RANDOM NETWORKS

## Random networks with different parameters

$p = 0{,}17; N = 12$



$p = 0{,}03; N = 100$



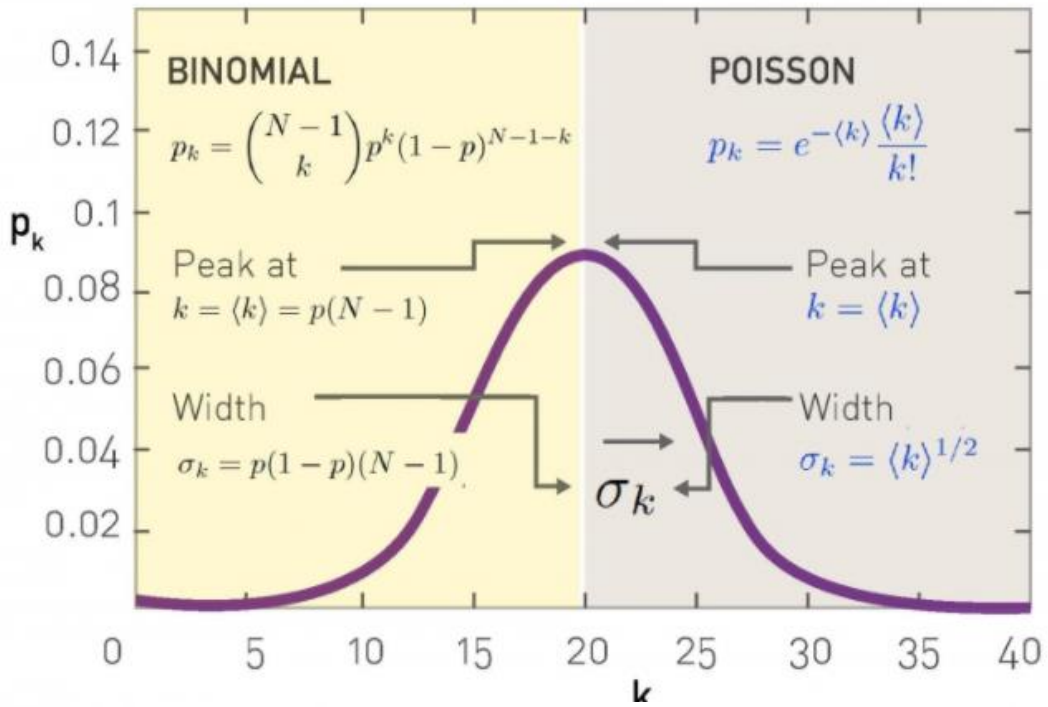
# RANDOM NETWORKS

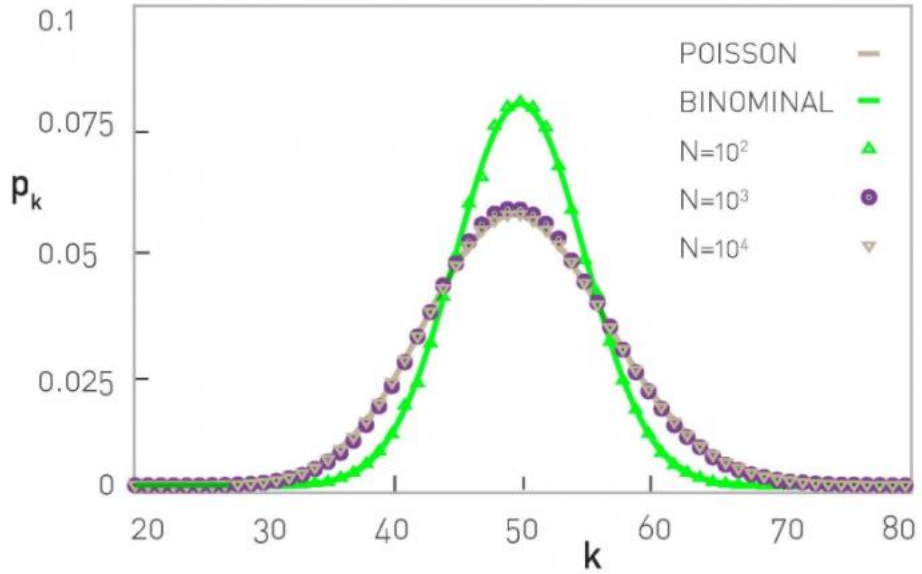
## The degree distribution of the random network

It' depend on three terms:

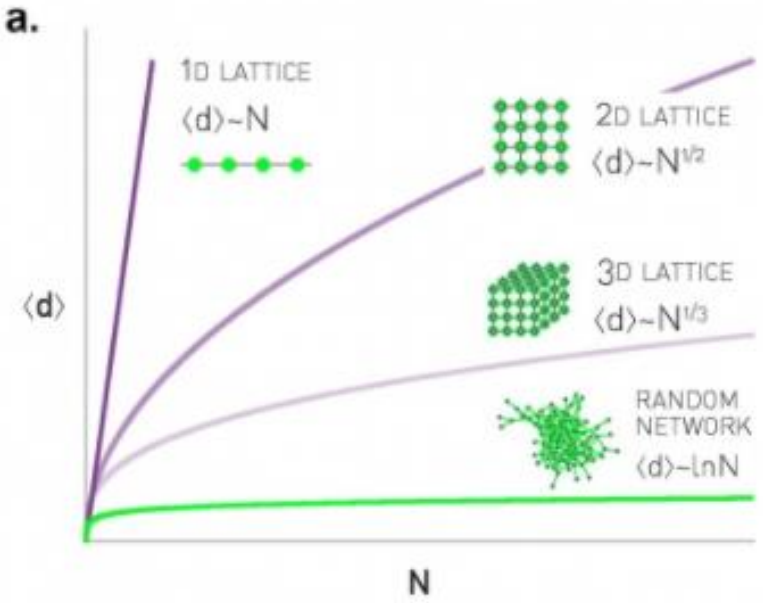
- the probability that *k* of its links are present,
- the probability that the remaining (*N-1-k*) links are missing,
- The number of ways we can select *k* links from *N- 1* potential links.

Binomial or Poisson distributios?



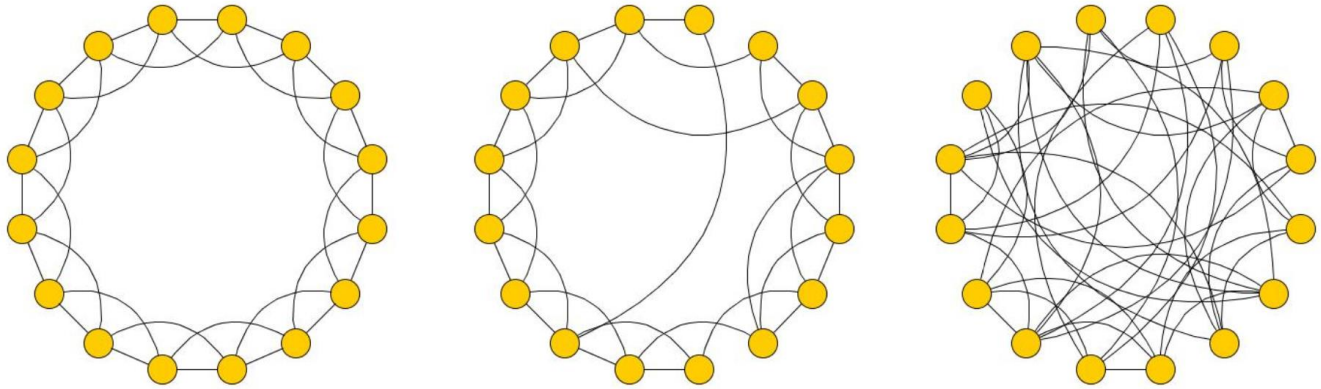
# SMALL WORLD NETWORKS




**But the big networks aren't Poisson type**

- In the random network missing the hubs.
- In the real networks the distances are short. (Six step separation)
- The small world property is defined by:

$$\langle d \rangle = \frac{lnN}{ln\langle k \rangle}$$

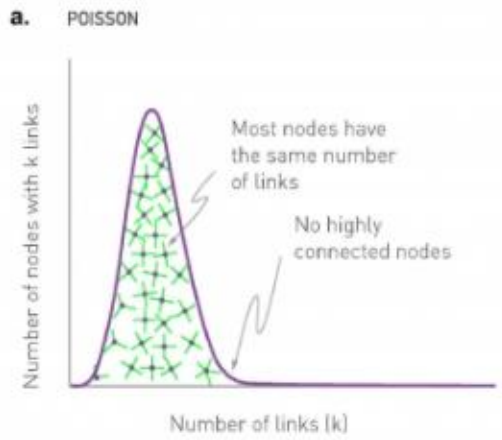


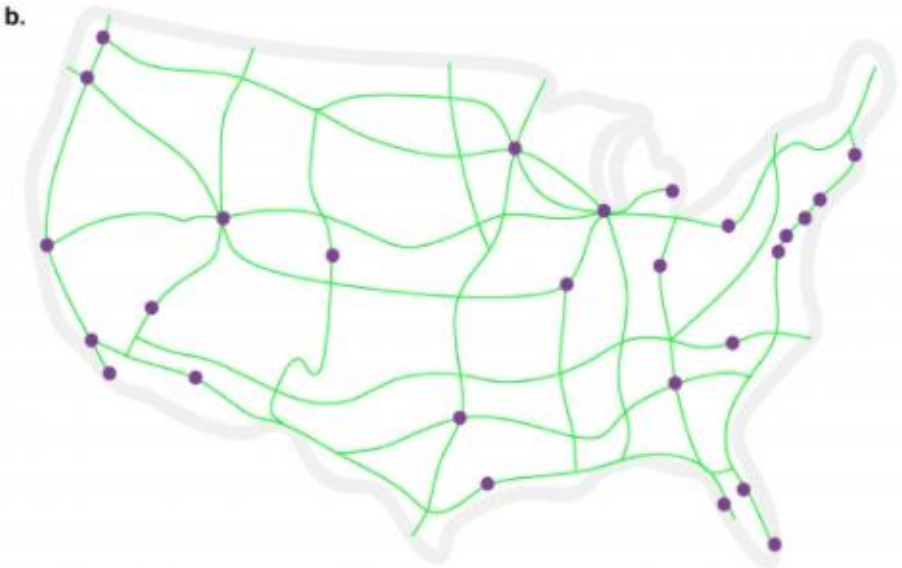
# SCALE FREE NETWORKS

## Random network

The degrees of a random network follow a Poisson distribution, rather similar to a bell curve.

Therefore most nodes have comparable degrees.

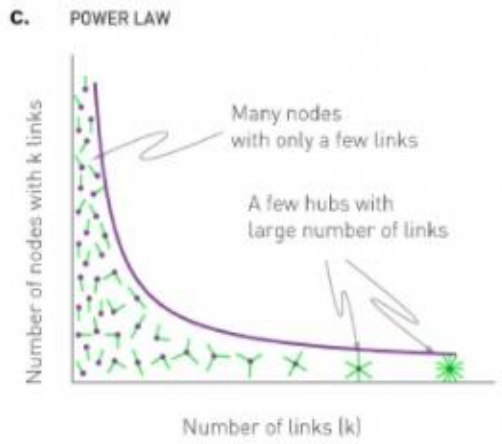



## Scale free network

In a network with a power-law degree distribution most nodes have only a few links.

These numerous small nodes are held together by a few highly connected hubs.






# DIFFERENT NETWORK

| Network | ⟨k⟩ | ⟨d⟩ | $d_{max}$ |
|---|---|---|---|
| Internet | | | |
| WWW | | | |
| Power Grid | | | |
| Mobile-Phone Calls | | | |
| Email | | | |
| Science Collaboration | | | |
| Actor Network | | | |
| Citation Network | | | |
| E. Coli Metabolism | | | |
| Protein Interactions | | | |

# THE BARABÁSI-ALBERT NETWORKS

**Random network**

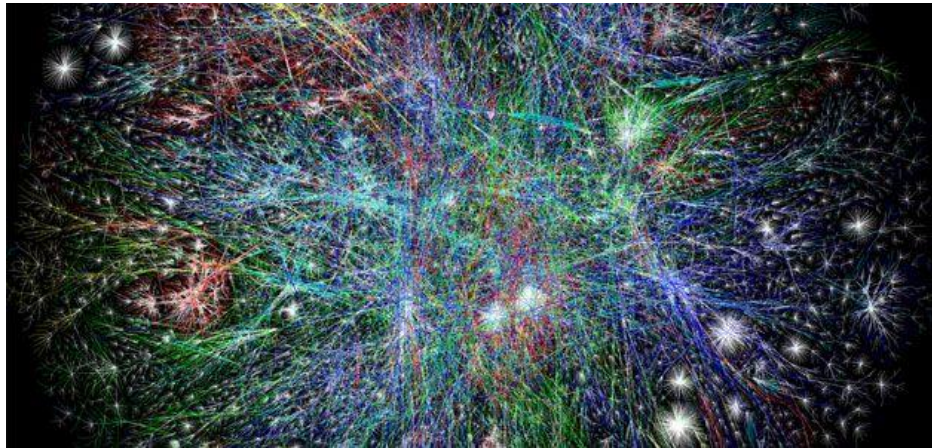
The random network model assumes that we have a *fixed* number of nodes, *N*.

It's a static model.

**Barabási-Albert network**

But, in real networks the number of nodes continually *grows* thanks to the addition of new nodes.

It's a dynamic model.



# THE BARABÁSI-ALBERT NETWORKS

**Growth**

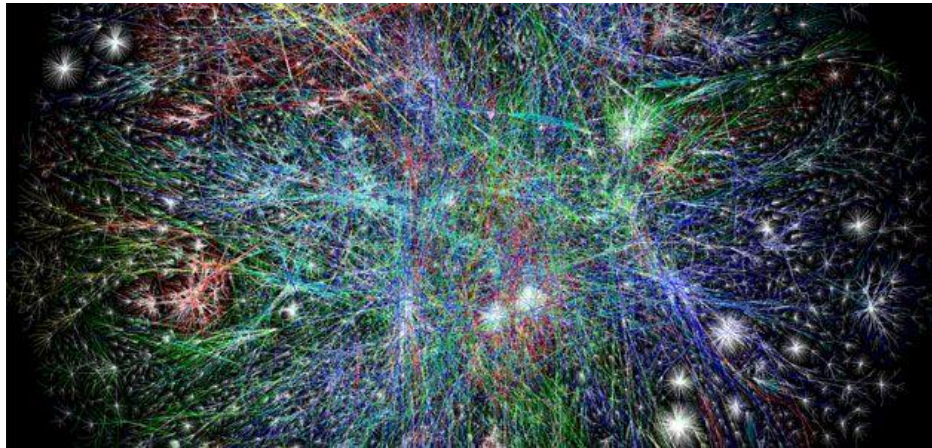
Real networks are the result of a growth process that continuously increases N.

In contrast the random network model assumes that the number of nodes, N, is fixed.

**Preferential Attachment**

In real networks new nodes tend to link to the more connected nodes.

In contrast nodes in random networks randomly choose their interaction partners.



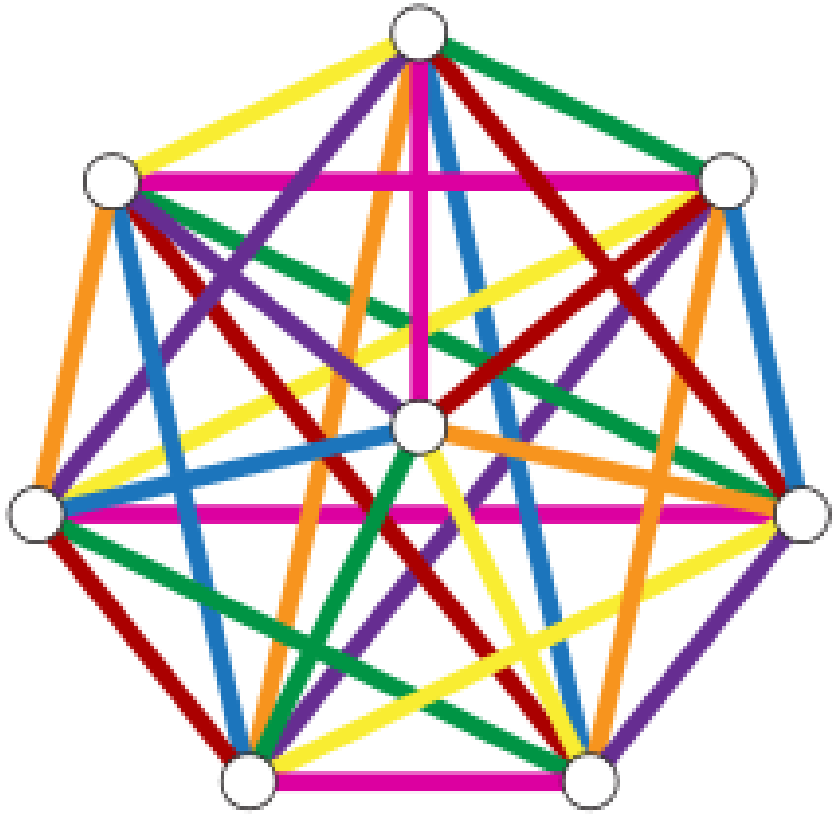
# THE MULTIPLEX NETWORKS

**simple graph:**

- there are only 1 connection
  between nodes

**colored graph:**

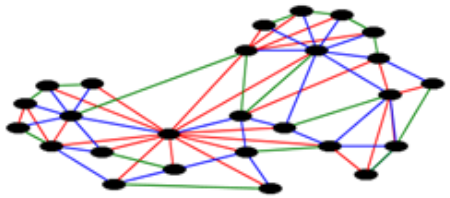
- would be **more than 1 connection**
  between nodes
- the connections are different types



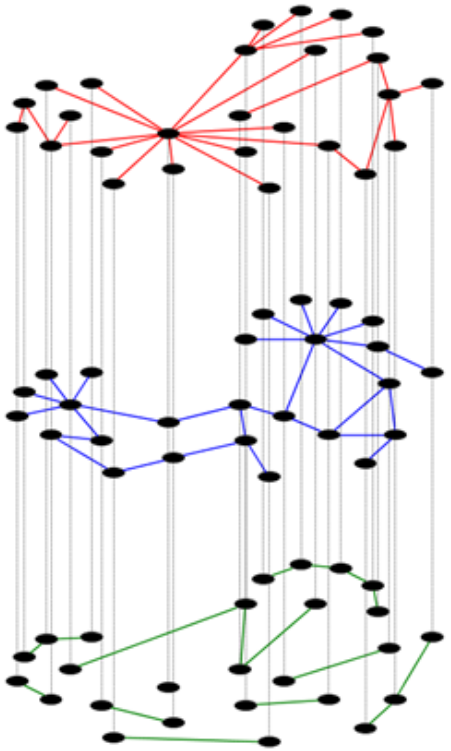
# THE MULTIPLEX NETWORKS

The colored graph has same colored subgraphs.

These subgraphs are the homomorph mapping of the original graph.



(a)

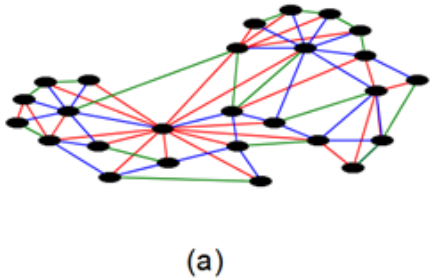


(b)

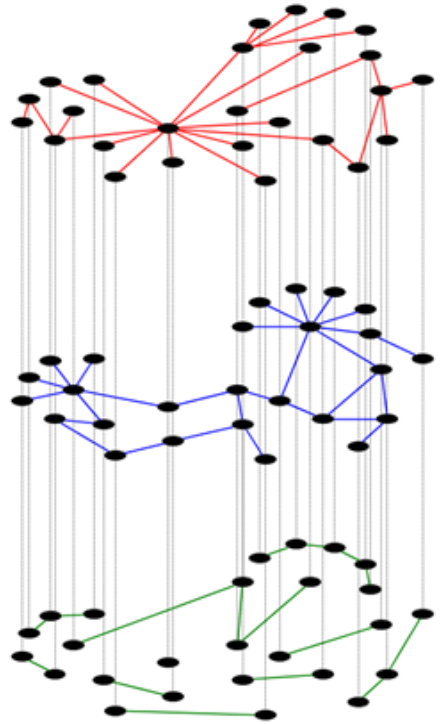
# THE MULTIPLEX NETWORKS

## DIFFERENT NETWORKS IN AN ORGANIZATION

- Personal power (hierarchical position)
- Professional ordering (professional position)
- Knowledge sharing(s) (formal/vouluntary)
- Substitutional(s) (real/ideal/potential)
- Processes
- Spatial
- Personal preferences
- Gossip
- etc.



(a)



(b)

# THE MULTIPLEX NETWORKS

They use some different phrases for the network that has more than one connection between the nodes.

The Kivela type terminology:

- **monoplex network** or **single-layer network:**

  There would be only one connection between two nodes.

- **interconnected network:**

  In the network are more than one monoplex networks and would be connections between the nodes of different monoplex networks.

- **network layer:**

  are the monoplex networks of the interconnected network.

# THE MULTIPLEX NETWORKS

- **intra-layer connections:**

  are the inside connections in a network layer.

- **inter-layer connections:**

  are the connections between the nodes of the different network layers.

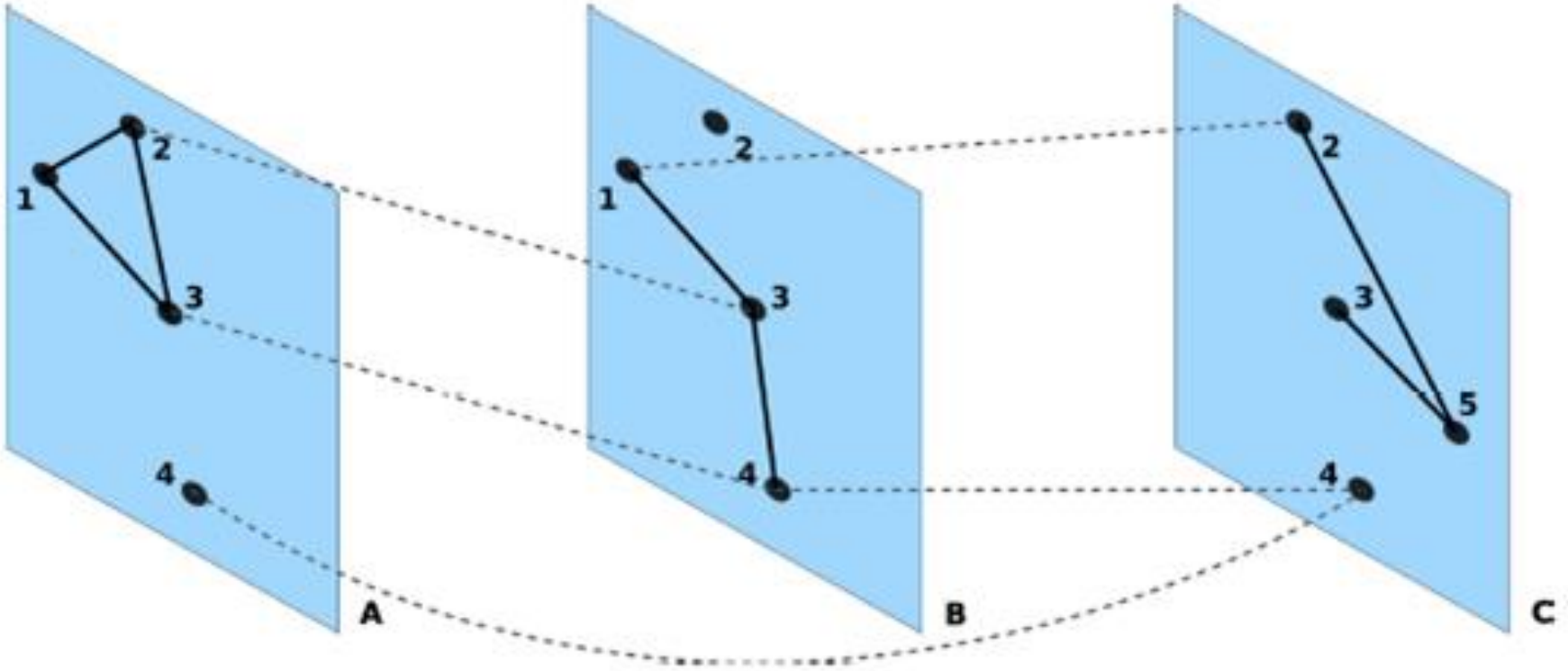
- **multilayer network:**

  is an interconnected network that the nodes of the network layers are same and would be inter-layer connection between any nodes.

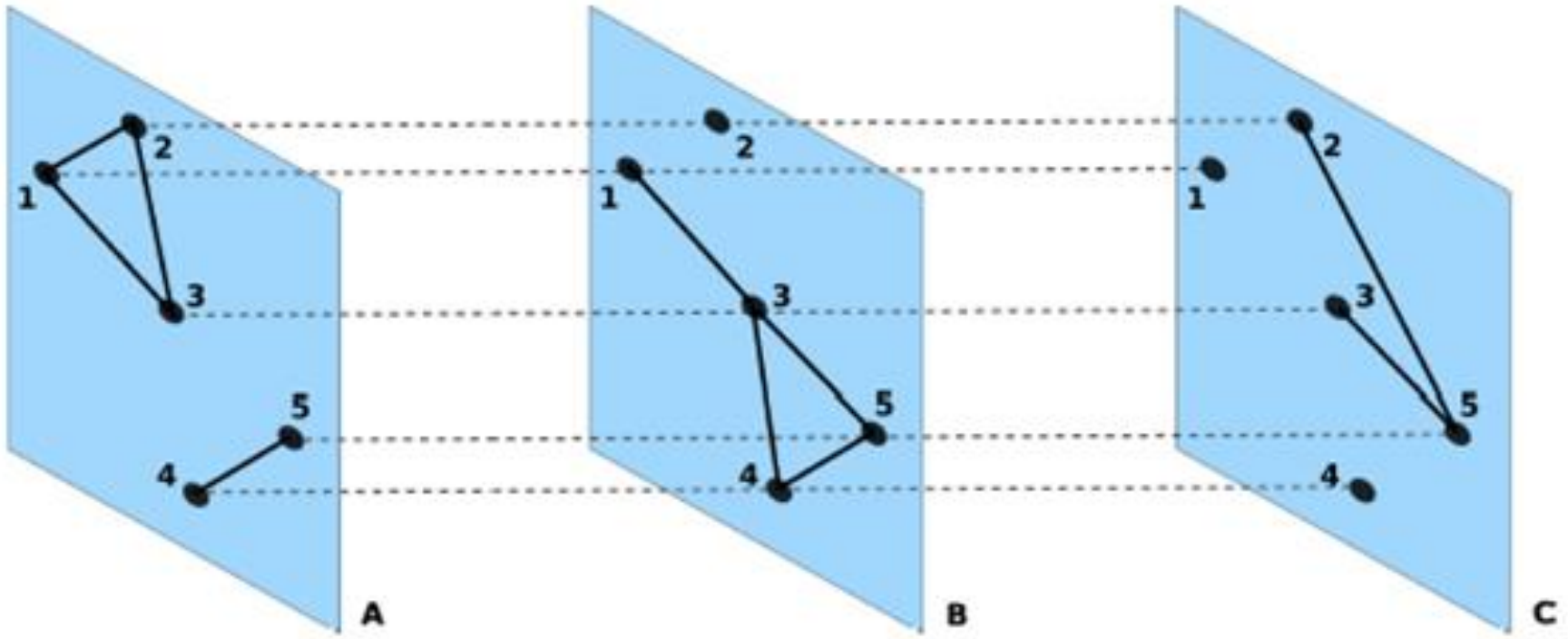
- **multiplex network:**

  is an interconnected network that the nodes of the network layers are same, but the inter-layer connection would be between same nodes only.

# MULTILAYER NETWORK (EXAMPLE)



# MULTIPLEX NETWORK (EXAMPLE)



# INFORMATION SPREADING IN THE ORGANIZATIONAL NETWORKS

- The organization is a complex human network.
- There are many network layers in an organization.
- Undirected and directed information layers.
- On every layers spread information.
- The different network layers have different topology.
- The information spreading depend on network topology.

# INFORMATION SPREADING IN THE ORGANIZATIONAL NETWORKS

**The parts of the information spreading model:**

- the expanded adjacency matrix $P$ of the network
- the initial $S_t$ state vector of the nodes

$$K + E = P$$

$$\begin{pmatrix} k_{11} & k_{12} & . & . & k_{1j} \\ k_{21} & k_{22} & . & . & . \\ . & . & . & . & . \\ . & . & . & . & . \\ k_{i1} & . & . & . & k_{ij} \end{pmatrix} + E = \begin{pmatrix} p_{11} & p_{12} & . & . & p_{1j} \\ p_{21} & p_{22} & . & . & . \\ . & . & . & . & . \\ . & . & . & . & . \\ p_{i1} & . & . & . & p_{ij} \end{pmatrix}$$

$$S_t := \begin{pmatrix} s_1 \\ s_2 \\ . \\ . \\ s_n \end{pmatrix}$$

$$s_n = \begin{cases} 1 \text{ if the person } n \text{ got the news} \\ 0 \text{ if the person } n \text{ didn't got the news} \end{cases}$$

# INFORMATION SPREADING IN THE ORGANIZATIONAL NETWORKS

**The information spreading model:**

$$P \times S_t \rightarrow S_{t+1}$$

$$\begin{pmatrix} p_{11} & p_{12} & \cdot & \cdot & p_{1j} \\ p_{21} & p_{22} & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ p_{i1} & \cdot & \cdot & \cdot & p_{ij} \end{pmatrix} \times \begin{pmatrix} s_1 \\ s_2 \\ \cdot \\ \cdot \\ s_n \end{pmatrix}_t = \begin{pmatrix} s_1 \\ s_2 \\ \cdot \\ \cdot \\ s_n \end{pmatrix}_{t+1}$$

Must iteration for $S_t = S_{t+1}$

# NETWORK ROBUSTNESS

## Robustness

- Errors and failures can corrupt all human designs.
- In this event you must call a profession.
- But many natural and social systems can sustain their basic functions even when some of their components fail.
- Networks play a key role in the robustness of biological, social and technological systems.
- What is the origin of the robustness in a network?

# NETWORK ROBUSTNESS

## Percolation Theory

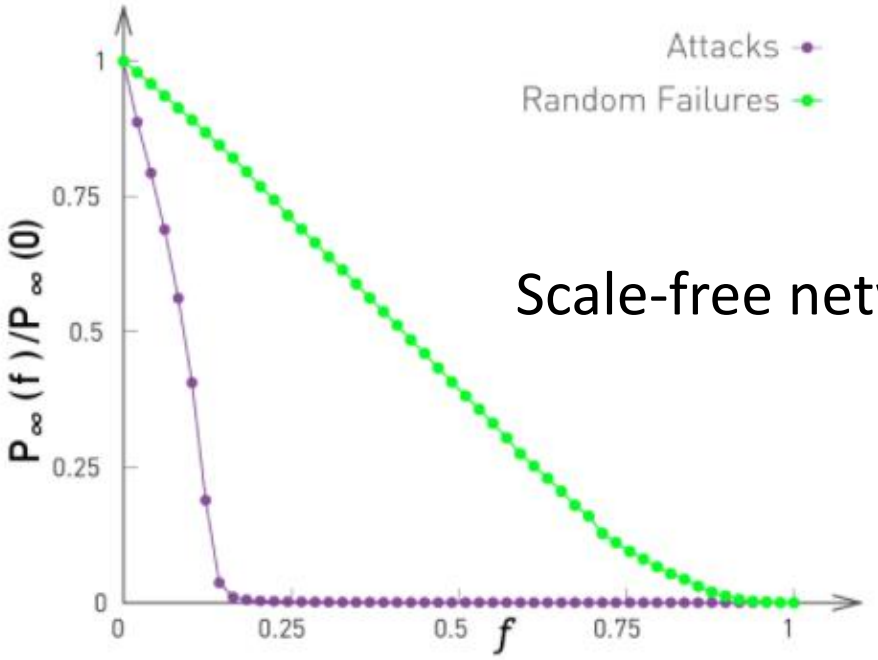
- Remove a node from the network!
- The removal of a single node only limited impact on network's integrity.
- The removal of several nodes, however, can break a network into several isolated components.
- How many nodes do we have to delete to fragment a network into isolated components?

# NETWORK ROBUSTNESS

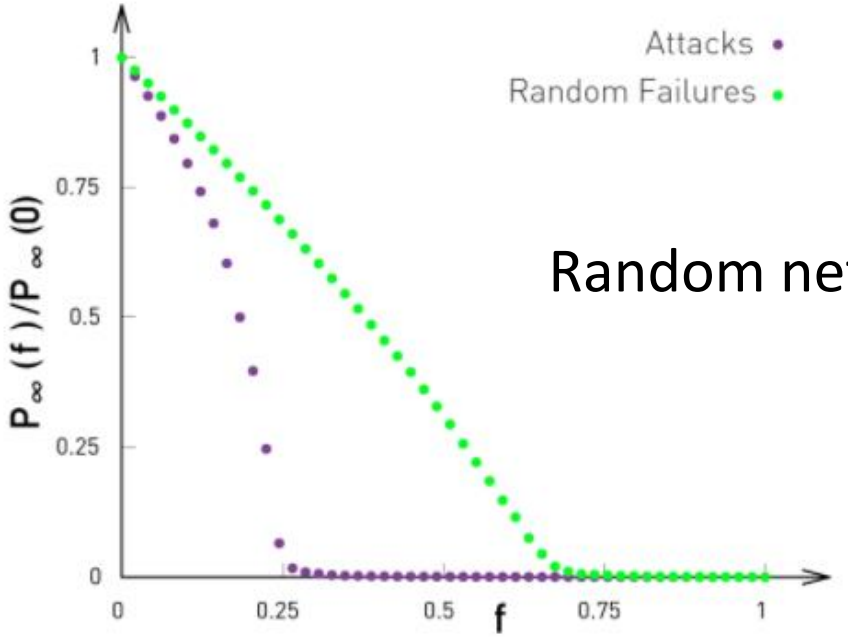
**Failure:** randomly error in the network.

**Attack:** intentional error in the network.

**The robustness depend on network topology.**




Scale-free network




Random network

# POWER NETWORKS IN THE ORGANIZATION

**Mastenbroek-type power structures**



Willem Frans Gerard Mastenbroek
(1942 - )
Dutch sociologist

Conflict Management and Organization Development (1988)
Negitiating – As Emotion Management (2012)

# POWER NETWORKS IN THE ORGANIZATION

## Mastenbroek type power structures

- **Instrumental relationships network**
  - In this network **the persons in relations of each other are considered as means of production**.
  - Since the performance of professional activities requires the co-worker's activity as well, this results the division of labor and coordination.
  - **The professional information flows** in this instrumental network, there is the provision of data, **the communication and the decisions** are made in this network as well.

# POWER NETWORKS IN THE ORGANIZATION

## Mastenbroek type power structures

- **Socio-emotional relationships network**
  - This network is based on the **emotional relationships** of co-workers structured **on the basis of sympathy and dislike**.
  - Key principles are belonging to the group and through that the **loyalty**, the elements of group cohesiveness such as **common language or group image**.
  - The **individual norms** may be set up in the socio-emotional networks.
  - The **emotions** dominate this emotional network.

# POWER NETWORKS IN THE ORGANIZATION

## Mastenbroek type power structures

- **Power and dependency network**
  - This is the place, where **organizational power game** is being played, inner **political manoeuvres** take place here, and **people fight for their positions and resources.**
  - These power games typically take place **in the background**, participants often use the strategy of misrepresentation, and **their individual intentions are covered under the organization interest**.

# POWER NETWORKS IN THE ORGANIZATION

## Mastenbroek type power structures

- **Negotiation relation network**
  - Where the organization appears as an **inner market**,
  - individuals **compete for resources** in this network,
  - divide the inner market, and in order to receive **greater part of resources,**
  - and to be more dominant in decision-making **people tend to expand their own market borders** continuously at other's expense.

# POWER NETWORKS IN THE ORGANIZATION

## Mastenbroek type power structures

- **„One to three” approach**
  - The **power and dependency relationship is the most important,**
  - at the same time **the most predominant relation,**
  - and the **other three relation forms can be deduced from that one**.

# POWER NETWORKS IN THE ORGANIZATION

## Mastenbroek type power structures

Mastenbroek considers a relevant issue the **rate of dependence and independence** of individuals and organizational units in the organization that is determined by three factors:

- The level of **uncertainty**.
- The level of **substitution**.
- The level and form of **centrality**.

# POWER NETWORKS IN THE ORGANIZATION

## Mastenbroek type power structures

**The level of uncertainty:**

- Uncertainty in this case means that **how much information the individual or the organizational unit has for the future**.
- Those who have information about the future events that are **more likely to occur** it is an **advantage** and they may have an **influence** on the others' position in the organization.

# POWER NETWORKS IN THE ORGANIZATION

## Mastenbroek type power structures

**The level of substitution:**

- An individual or a group, that is difficult or impossible to substitute within an organization, may get considerable power by being indispensable.
- Hence it is a common behavior in the organization that the individuals or the groups tend **to be indispensable for the organization**.

# POWER NETWORKS IN THE ORGANIZATION

## Mastenbroek type power structures

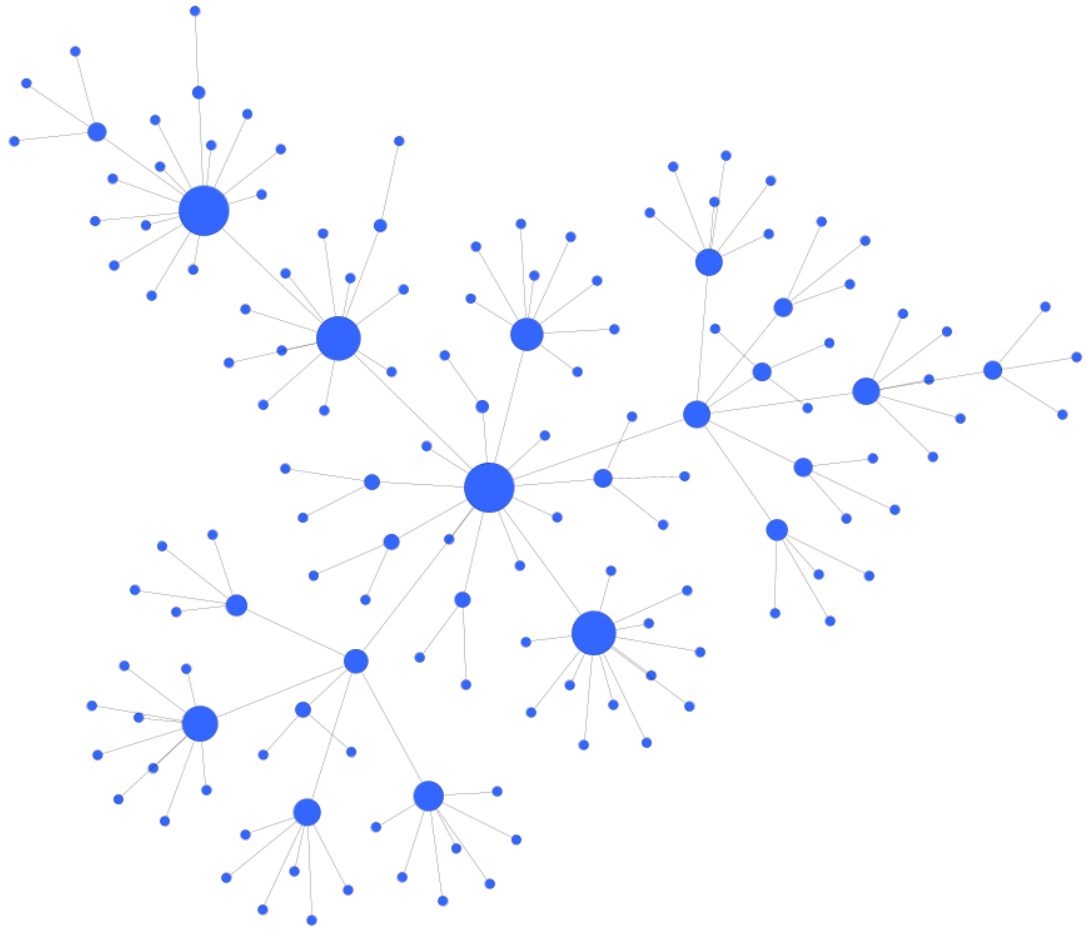
**The level and form of centrality:**

- **Interdependence** may be between the individuals and the groups.
- In case there is interdependence between the individuals and/or the groups, mutual dependence occurs and **results a strong interdependence between them**.
- The **greater importance** the person has in the organization, the **bigger power** he or she might have.

# POWER NETWORKS IN THE ORGANIZATION

## The Power and Dependency Network

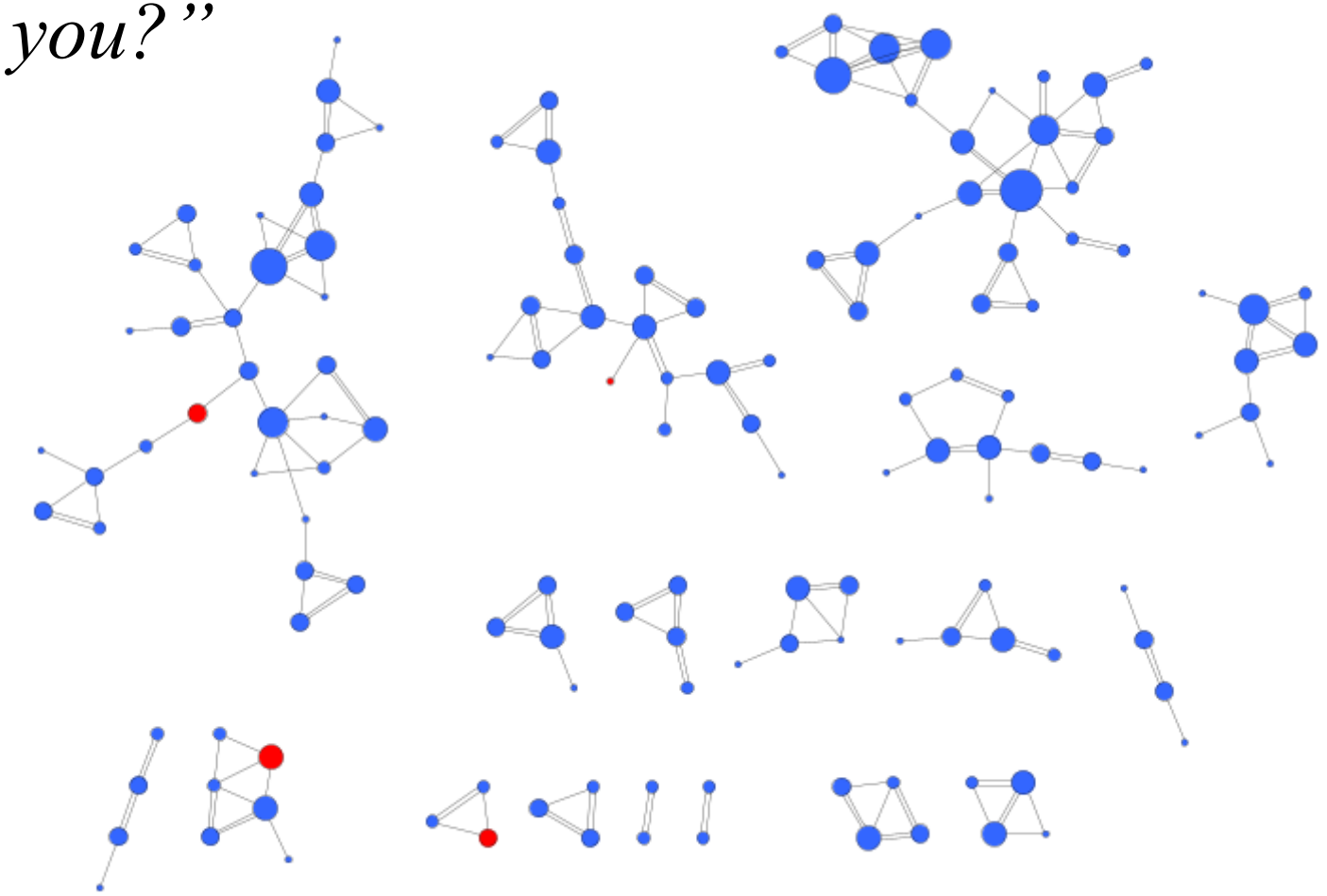
*"Who is/are your direct superior(s)?"*



# POWER NETWORKS IN THE ORGANIZATION

## The Power and Dependency Network

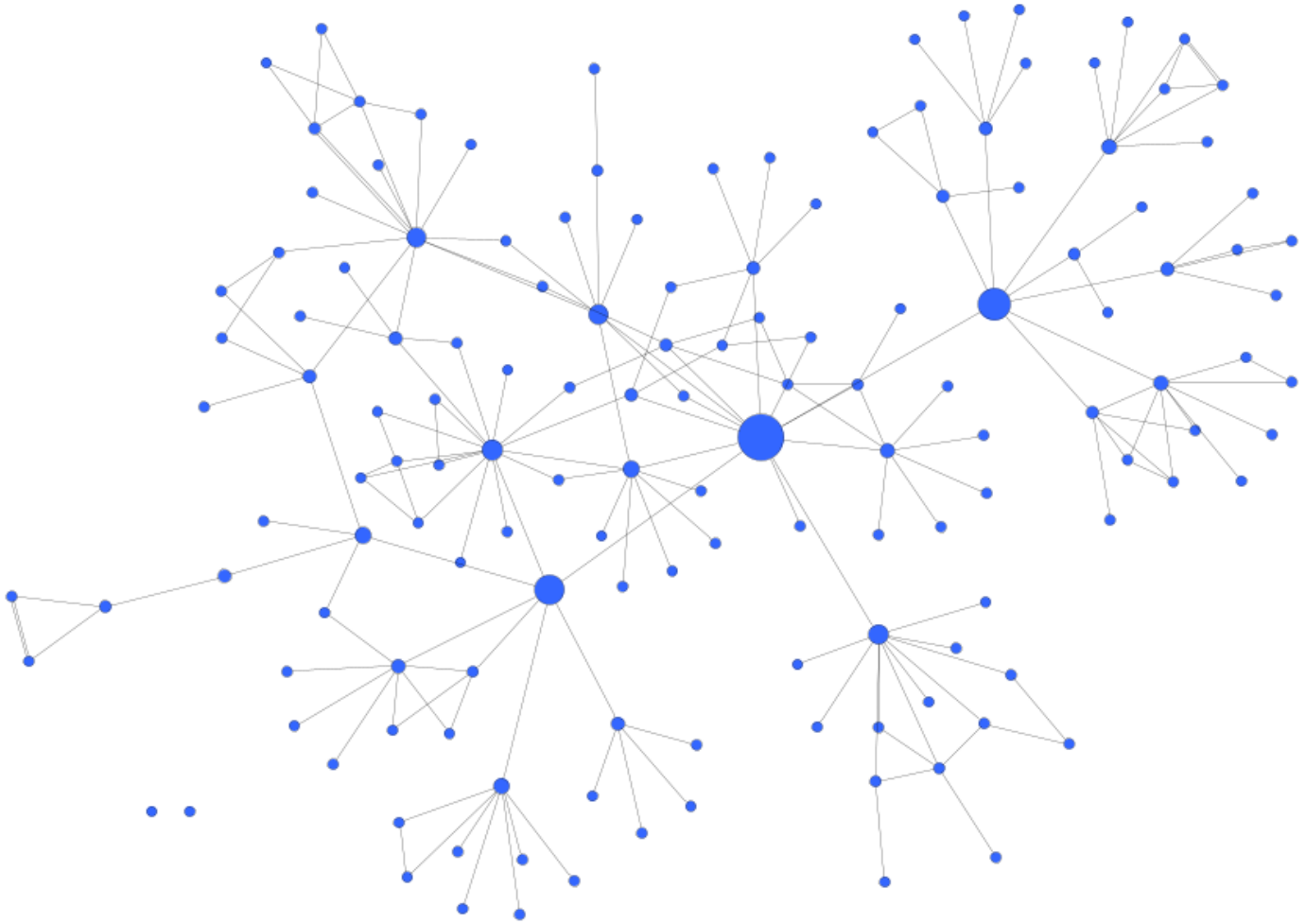
*"Who does usually substitute you?"*



# POWER NETWORKS IN THE ORGANIZATION

## The Negotiation Relation Network

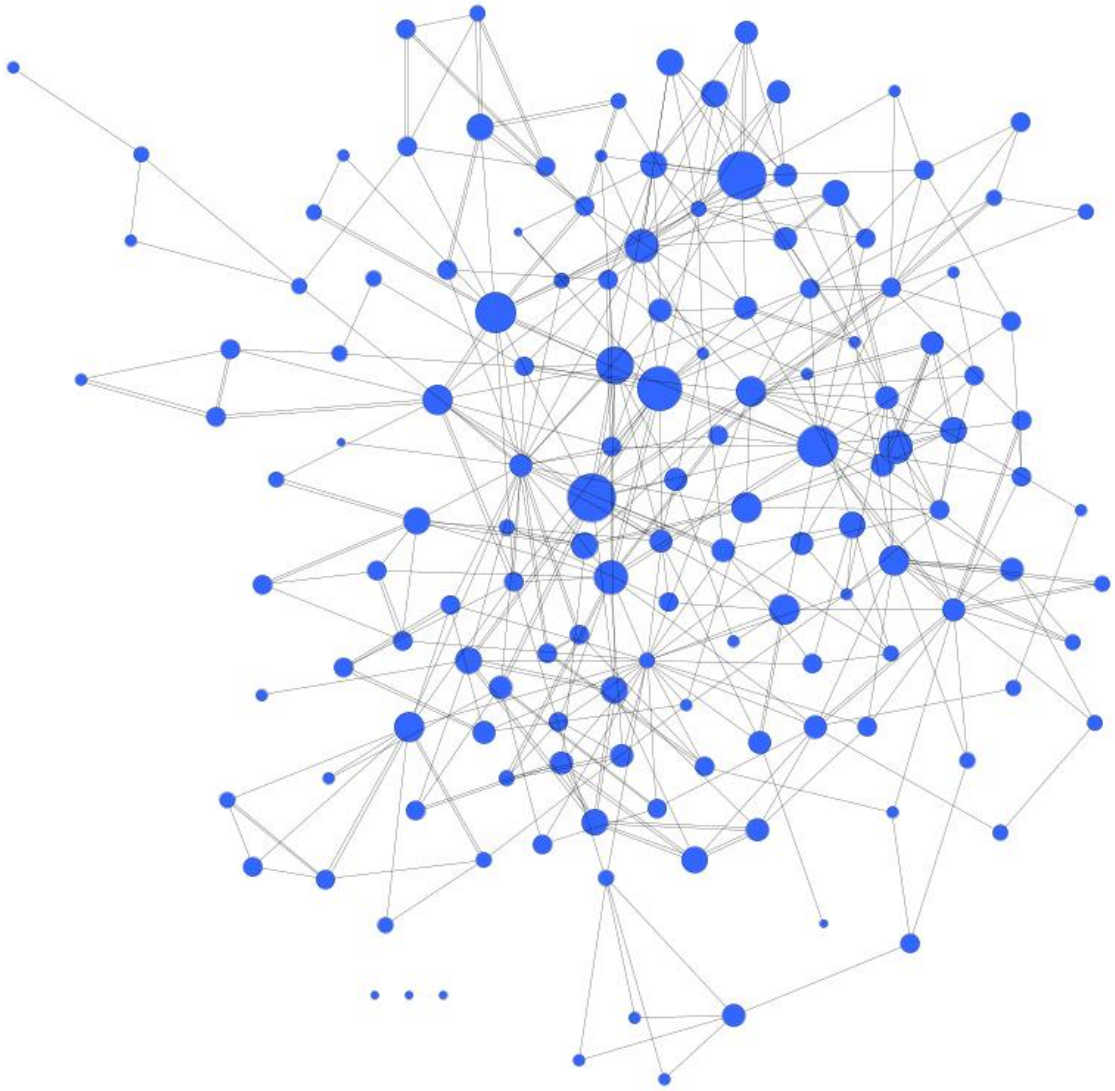
*"Who are you able to give professional instructions?"*



# POWER NETWORKS IN THE ORGANIZATION

## The Instrumental Network

*"Who do you usually provide expert information?"*



# POWER NETWORKS IN THE ORGANIZATION

## The Socio-Emotional Network

*"Who do you usually talk about non-professional issues on daily basis?"*

